

Digital Transfromation

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Chapter 1. Understanding Digital Transformation (What is Digital Transformation ?)



Business, Technology, Data and People

Let's start by establishing the overall purpose and scope of digital transformation. In this chapter we learn how business, technology, data and people must all align for a harmonious transformation.

Please prepare yourself.

We are about to... "transform".



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A digital transformation initiative is a genuine attempt to change, upgrade and extend an organization's business models and technologies so as to enable the organization to:

- ❑ gain value by significantly improving what it has been doing
- ❑ gain value by introducing new things it can do (and to ensure it can do those new things really well)

A digital transformation can introduce:

- ☐ a new mindset
- ☐ a new culture
- ☐ a new organizational structure
- ☐ new priorities
- ☐ new technologies
- ☐ new risks and challenges

A digital transformation will impact an organization's:

- ☐ business models and processes
- ☐ automation and data science technologies
- ☐ human workers and culture

- ❑ Specifically, the mainstream availability of a specific set of technology innovations and practices has enabled organizations to transform their businesses through enhanced automation and data intelligence to achieve greater business value and growth.
- ❑ This transformation results in changes to organizational culture and the positioning and allocation of human workers.
- ❑ The upcoming chapters explore the business and technology drivers that led to the emergence of mainstream digital transformation, as well as the common goals, benefits, risks and challenges of digitally transforming an organization.
- ❑ But first, this chapter takes a closer look at what fundamentally comprises digital transformation.

Business, Technology, Data and People

Although technology innovations are responsible for enabling digital transformation, the motivations, impacts, and influences of an organization's digital transformation effort go well beyond technology ([Figure 1.1](#)).

As explained in the upcoming sections, these factors involve the organization's business, its data and its people.



Figure 1.1

A digital transformation involves four primary factors: business, technology, data and people.

Digital Transformation and Business

Digital transformation is a genuine attempt to take an organization's business to a new level of growth and effectiveness by transforming:

- ❑ how the business relates to the outside world (in particular, how customers perceive and inter-relate with the business), and...
- ❑ how the business operates internally (in particular, how different organizational departments may need to collaborate, and how humans and machines may need to collaborate and inter-relate with each other).



When successfully realized, digital transformation can significantly reshape and enhance many aspects of an organization's business, including:

- ☐ the extent to which the business can make existing products and services available to new customers and new markets
- ☐ the extent to which the business can introduce new products and services into new markets
- ☐ the speed at which the business can penetrate and develop new markets
- ☐ the extent to which the business can leverage data
- ☐ the experience that customers have when interacting with the business
- ☐ the efficiency with which the business can operate and the level of automation the business can incorporate
- ☐ the agility with which the business can maneuver and adapt to change
- ☐ the manner in which the business can collaborate with partners



A common, overarching strategic goal of digital business transformation is to instill customer-centricity as being paramount to business models, automation and the organizational culture.

A successful digital business transformation (including the attainment of a meaningful extent of enhanced customer-centricity) is enabled by the successful application of contemporary technology and data intelligence, along with the successful contributions of human workers ([Figure 1.2](#)).

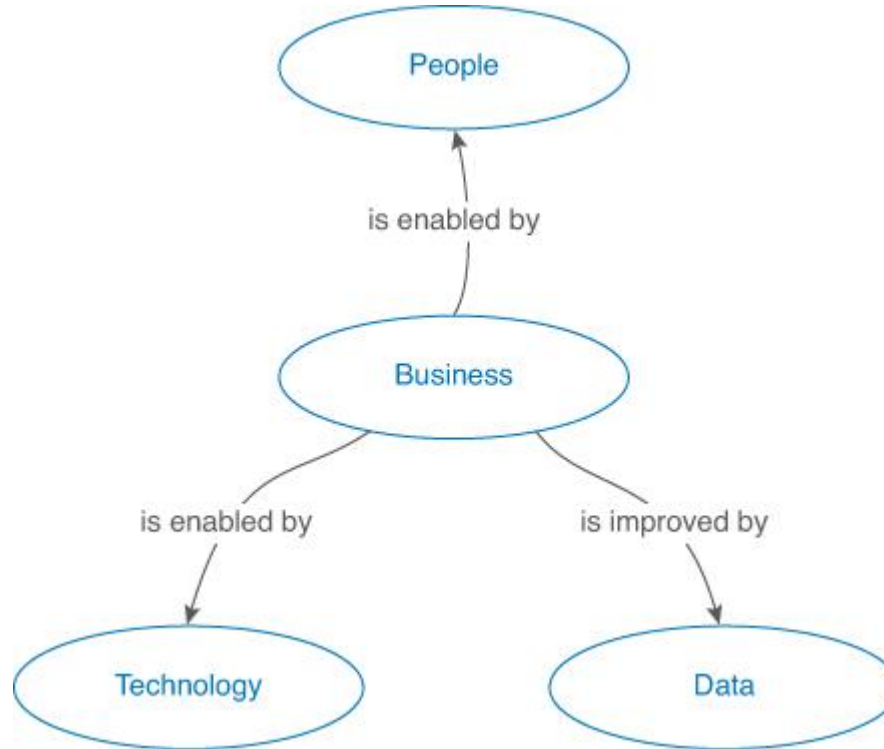


Figure 1.2

The business improvements and the introduction of new lines of business are enabled by human workers and automation technology, and further enhanced by increased data intelligence.

Digital Transformation and Technology

What has made digital transformation possible is a set of technology innovations that enable organizations to carry out significant and meaningful business transformations.

The primary distinguishing automation technologies are:

- ☐ Cloud Computing
- ☐ Blockchain
- ☐ Internet of Things (IoT)
- ☐ Robotic Process Automation (RPA)



The primary distinguishing data science technologies are:

- ☐ Big Data Analysis and Analytics
- ☐ Machine Learning
- ☐ Artificial Intelligence (AI)



Together, these technology innovations empower business capabilities in two primary areas:

- ❑ improved and new forms of automation
- ❑ more meaningful and responsive data intelligence
- ❑ An organization's business requirements and goals in relation to its planned digital transformation will help determine the correct combination of these technologies.

While the chapters in [Part I](#) of this book make occasional references to these technologies, much of [Part II](#) is dedicated to describing the technologies and exploring how they can be positioned as part of digital transformation solutions and platforms ([Figure 1.3](#)).

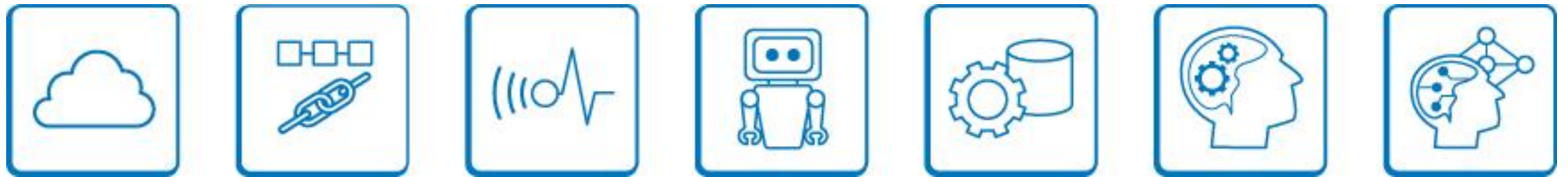


Figure 1.3

The icons used in the chapters in [Part II](#) of this book to represent the primary digital transformation technologies. From left to right: Cloud Computing, Blockchain, Internet of Things (IoT), Robotic Process Automation (RPA), Big Data, Machine Learning and Artificial Intelligence (AI).



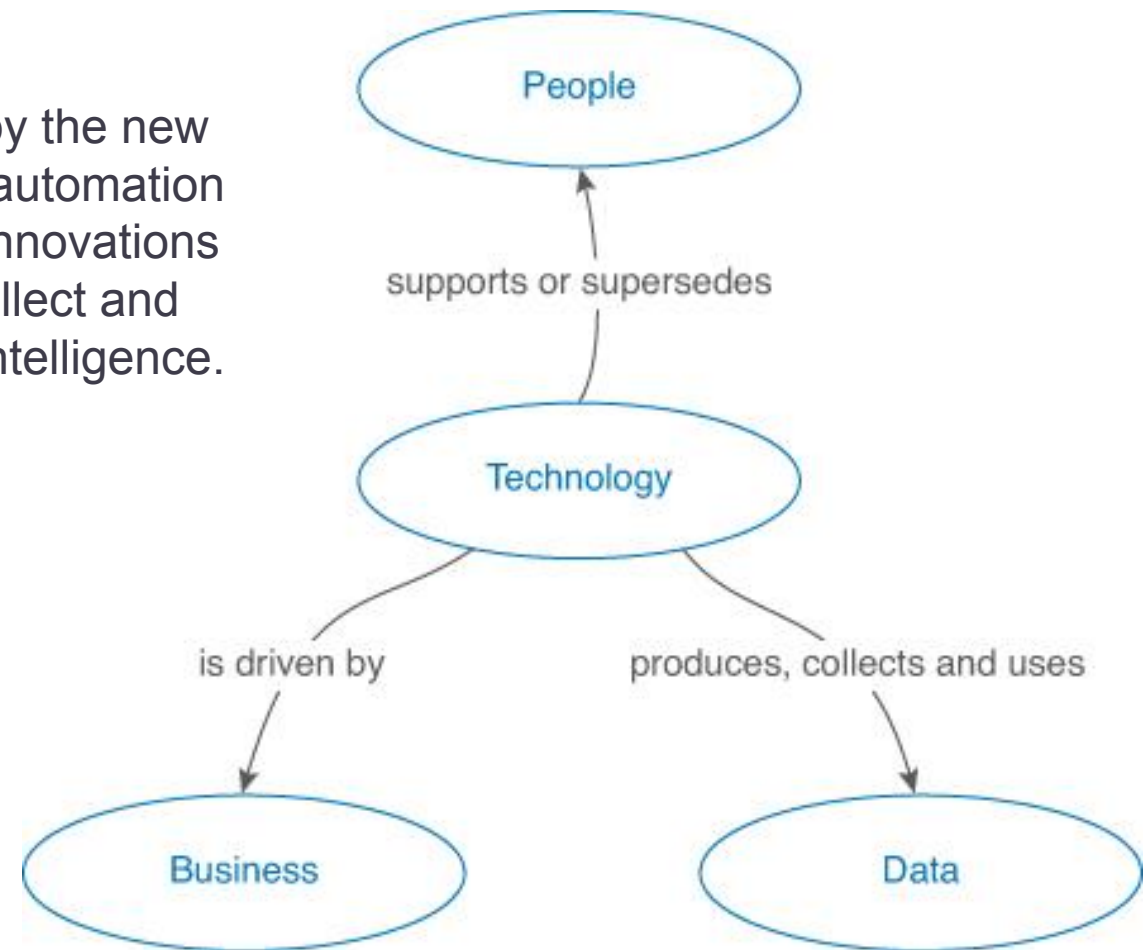
The utilization of these technologies is driven by the requirements of the business, and they correspondingly generate and process data in support of business data intelligence requirements.

Human workers can be further supported by the digital transformation solutions resulting from the application of these technologies.

In some cases, the new capabilities introduced by the solutions may allow manual tasks carried out by human workers to be automated by the solutions ([Figure 1.4](#)).

Figure 1.4

The utilization of technology innovations is determined by the new business requirements, which also determine how new automation technology relates to human workers. New technology innovations further introduce the capabilities necessary to create, collect and process new data in order to produce meaningful data intelligence.



Digital Transformation and Data

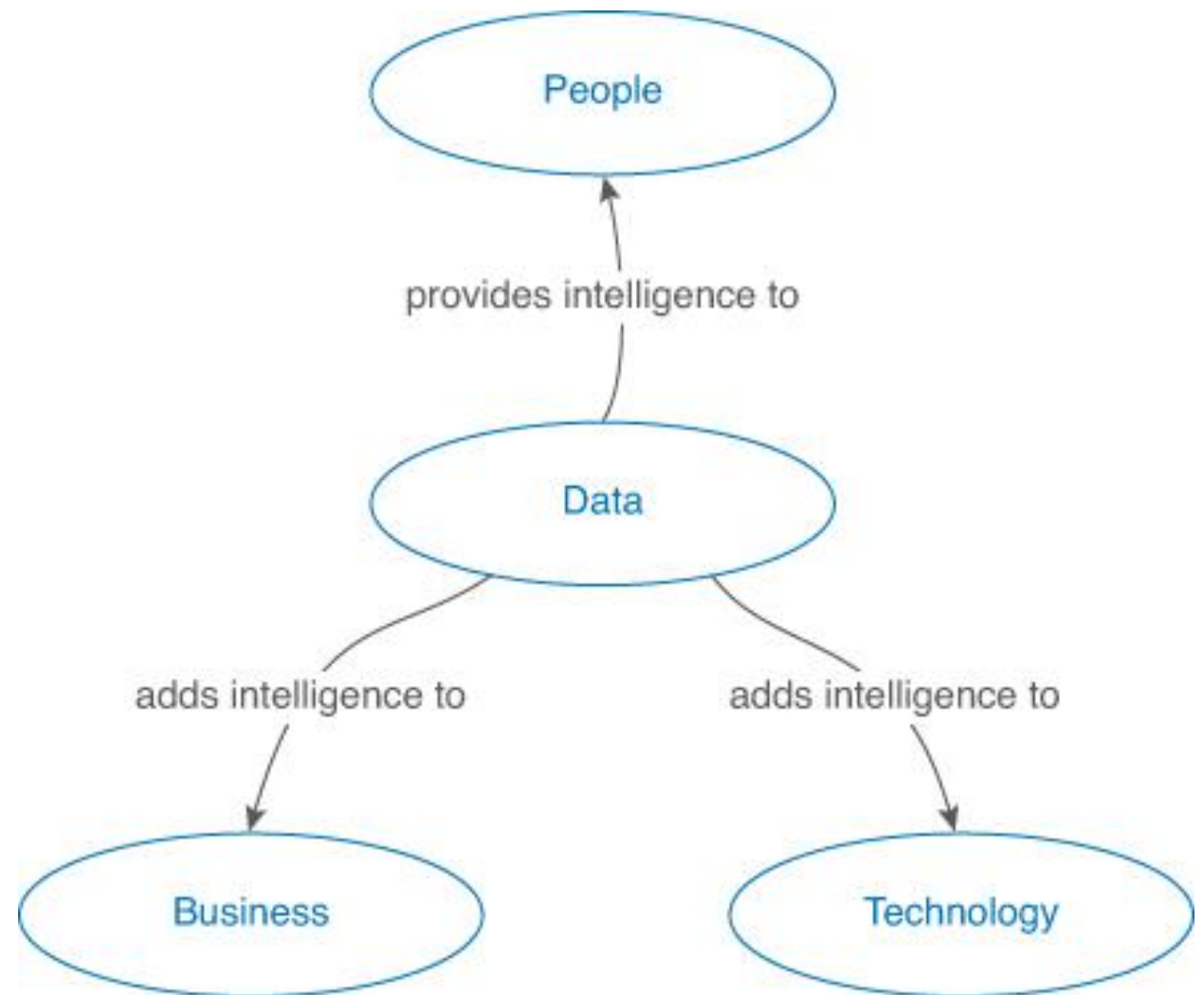
Digital transformation leads to automated business processes that are naturally data-driven. Data becomes a prime asset that empowers the organization to develop sophisticated business automation. The utilization of contemporary data science technologies as part of a digital transformation can inject an organization with large quantities of “data intelligence,” meaning the organization will have the opportunity to:

- ☐ gain meaningful insights into its own operations
- ☐ gain meaningful insights into its market or community
- ☐ gain meaningful insights into customer history and behavior
- ☐ gain meaningful insights into its existing products and services
- ☐ discover opportunities to introduce new products and services into its market or community
- ☐ enable its systems to learn from historical data to produce improved analysis results
- ☐ enable its systems to assume decision-making responsibilities

Data intelligence can be used to improve automation and increase opportunities to automate manual tasks previously carried out by human workers. Data intelligence can further support human workers responsible for carrying out decisions ([Figure 1.5](#)).

Figure 1.5

The infusion of deeply meaningful and insightful data enables automation solutions, human workers and the organization itself to carry out business more intelligently.



Digital Transformation and People

The transformation of both business and technology landscapes within an organization will naturally augment or reshape traditional organizational structures and change the way human workers contribute and relate to the organization.

For example:

- ❑ Sophisticated new technologies can automate a range of tasks that have historically been carried out manually by human workers.
- ❑ Human workers that previously performed menial tasks that are automated can now be retrained and reallocated to carry out more challenging and meaningful tasks.
- ❑ Human workers with new skill-sets are required to implement and maintain new automation technologies and, in particular, new data science technologies.
- ❑ The introduction of data intelligence as part of digital transformation solutions and as part of enhanced or new business operations can introduce the need for new knowledge and skill development, both in terms of producing and utilizing the data intelligence.
- ❑ Human workers affected by changes to business models and business processes may need to be retrained and perhaps even acquire new skills.
- ❑ The introduction of new products and services can introduce the need to hire new human workers capable of operating within the new supporting business processes.

All of these changes to how human workers carry out new or improved business tasks, work with new technology, learn from new data intelligence and collaborate and relate to the organization in new ways ([Figure 1.6](#)), can lead to the formation of new organizational cultures, more integrated with digital capabilities and focused on customer-centricity.

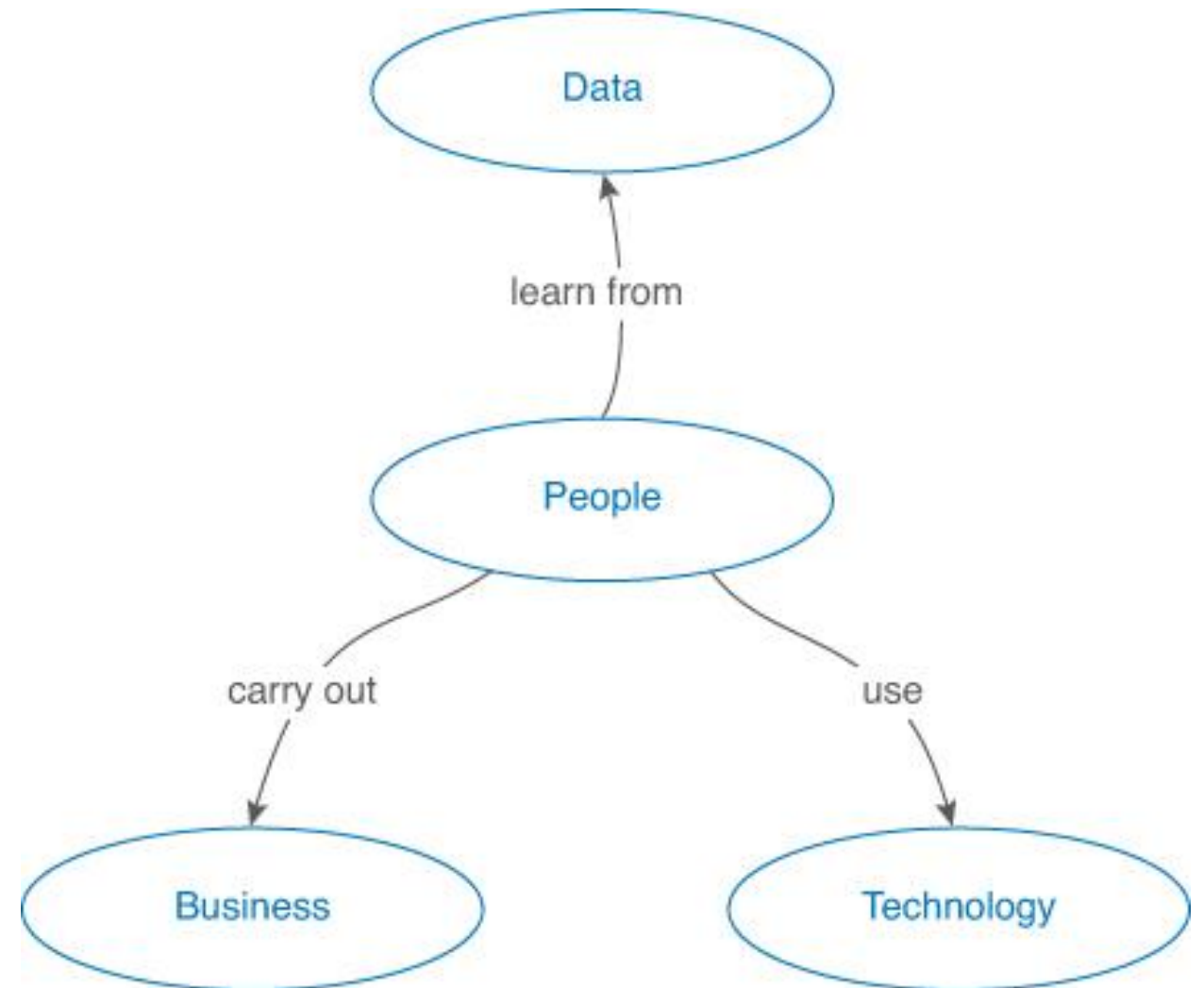
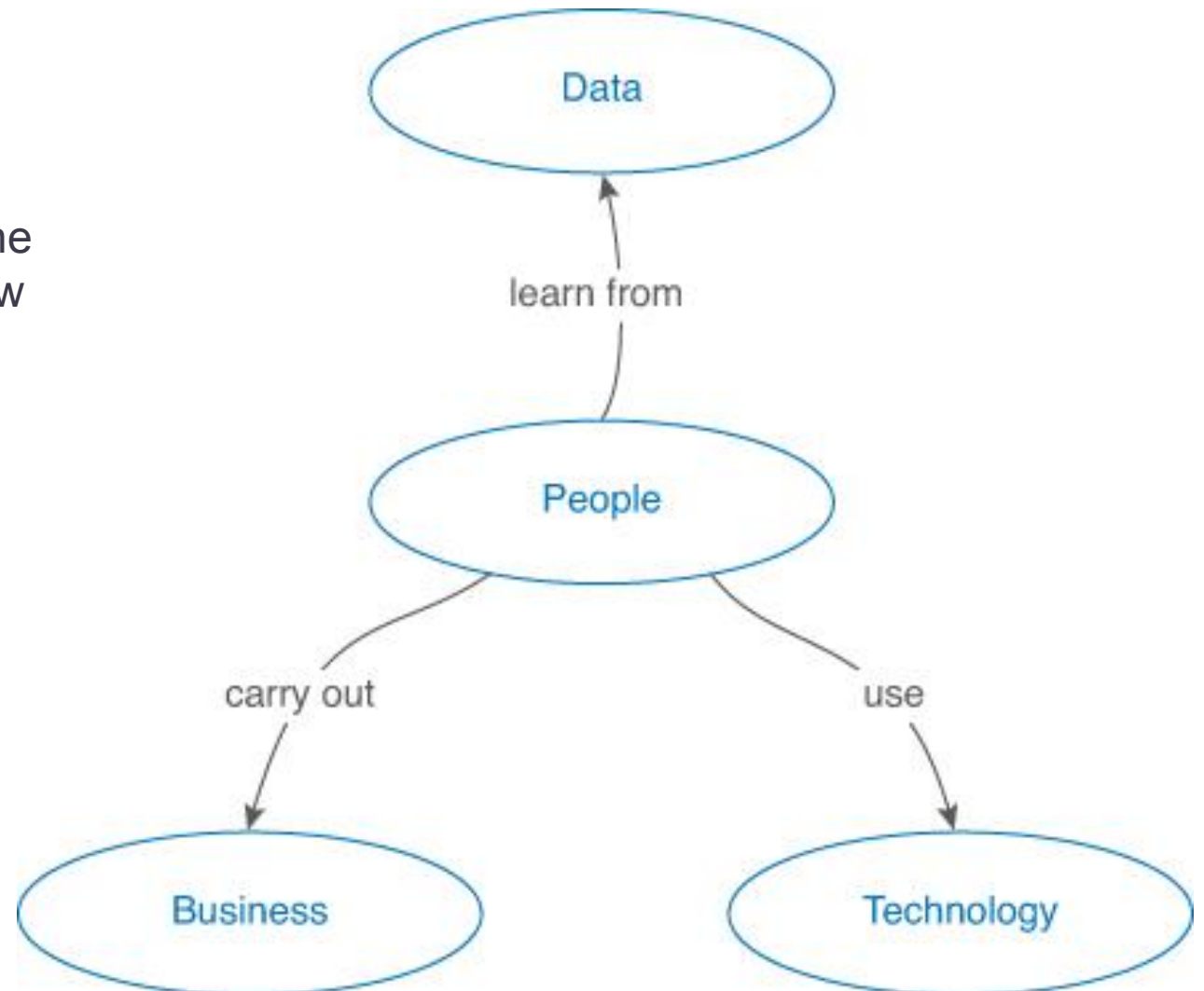


Figure 1.6

Human workers can be empowered to carry out the business more effectively via the availability of new automation technology and data intelligence.



Digital Transformation and Organizations and Solutions

As previously established, an organization undergoing a digital transformation will inevitably introduce benefits and impose challenges on its business operations, its technology ecosystem, the utilization of its data, as well as its relationships with its human workers ([Figure 1.7](#)).

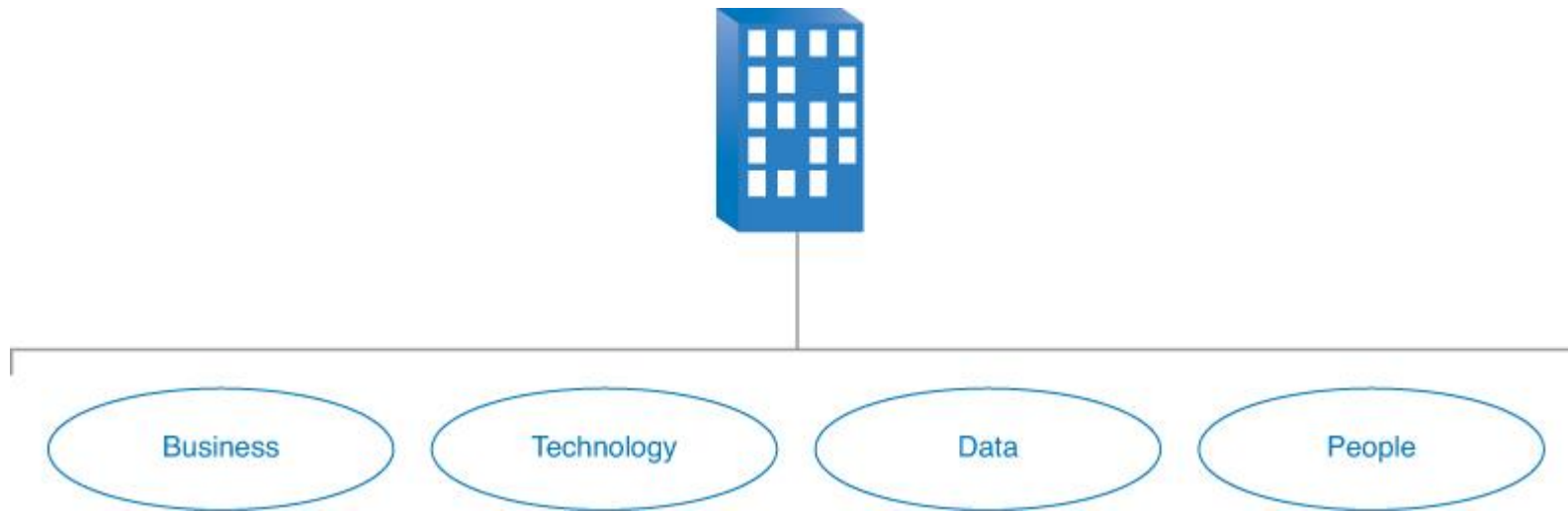


Figure 1.7 An organization's transformation relates to and impacts its business, technology, data and people.

To proceed with a digital transformation, the organization will need to develop *digital transformation solutions*, which are essentially applications responsible for automating and/or contributing data intelligence, each of which may encompass one or more related business tasks.

An organization will typically end up building multiple digital transformation solutions that share resources, such as infrastructure and databases as part of a greater platform ([Figure 1.8](#)).

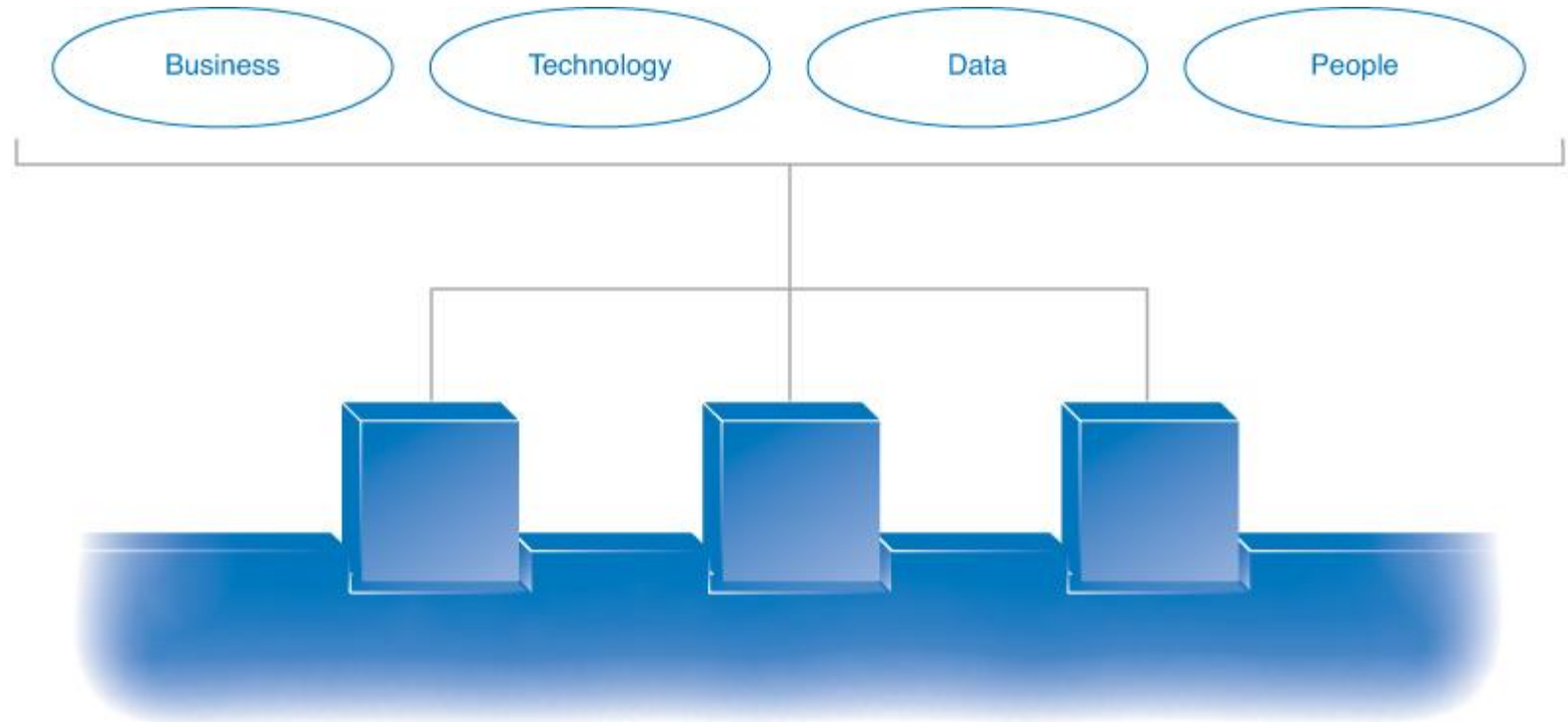


Figure 1.8

The introduction of digital transformation solutions enables an organization to carry out its new business transformation requirements via new technologies capable of producing and processing new data intelligence. Digital transformation solutions will further change how and where people within the organization contribute to business operations.

Chapter 2. Common Business Drivers (What Led to Digital Transformat ion?)



What Led to Digital Transformation?)

- Losing Touch with Customer Communities
- Inability to Grow in Stale Marketplaces
- Inability to Adapt to Rapidly Changing Marketplaces
- Cold Customer Relationships
- Inefficient Operations
- Inefficient Decision-Making





The best starting point to better understanding digital transformation is learning about the reasons that led to its emergence. This brief chapter covers key business drivers, each of which describes a common business problem that organizations have been facing and that is addressed by digital transformation.

The following key business drivers are covered:

- Losing Touch with Customer Communities
- Inability to Grow in Stale Marketplaces
- Inability to Adapt to Rapidly Changing Marketplaces
- Cold Customer Relationships
- Inefficient Operations
- Inefficient Decision-Making

Losing Touch with Customer Communities

Figure 2.1

Vendor A does not change its online applications as the years go by while Vendor B only changes its applications incrementally. Vendor C makes an effort to significantly upgrade and transform the products and services available via its online applications on a regular basis, thereby allowing it to lure more customers away from Vendors A and B.



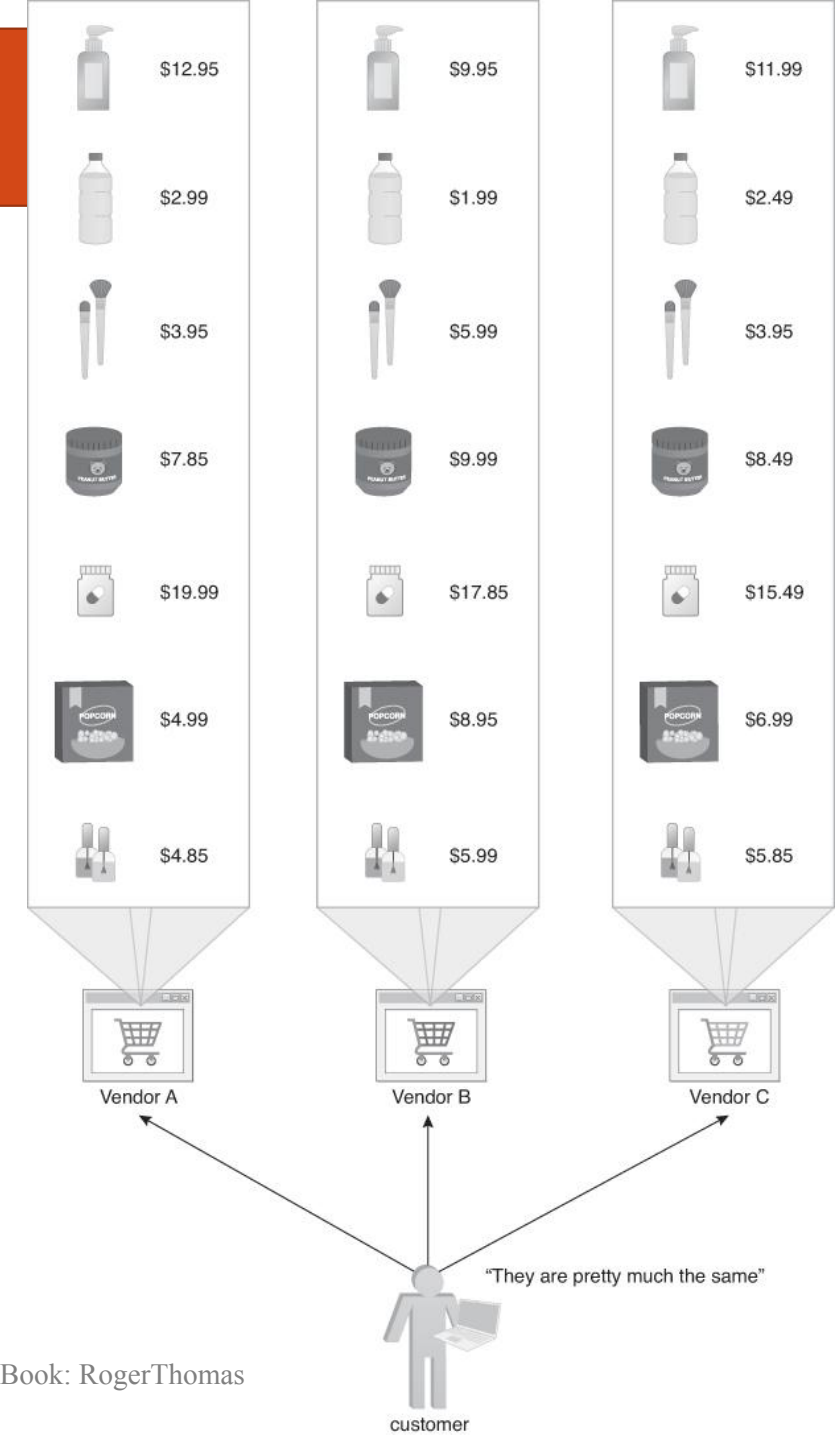
Inability to Grow in Stale Marketplaces

For years, several marketplaces did not change because businesses established themselves as the primary suppliers in those markets and, from a customer's perspective, there was little to distinguish one from the other ([Figure 2.2](#)).

This has stalled the growth of some organizations while completely inhibiting the growth of others.

Figure 2.2

The customer browses several online stores to decide from which it will purchase a specific set of grocery and beauty items. Other than price and shipping options, the customer sees little difference between what the online vendors have to offer.





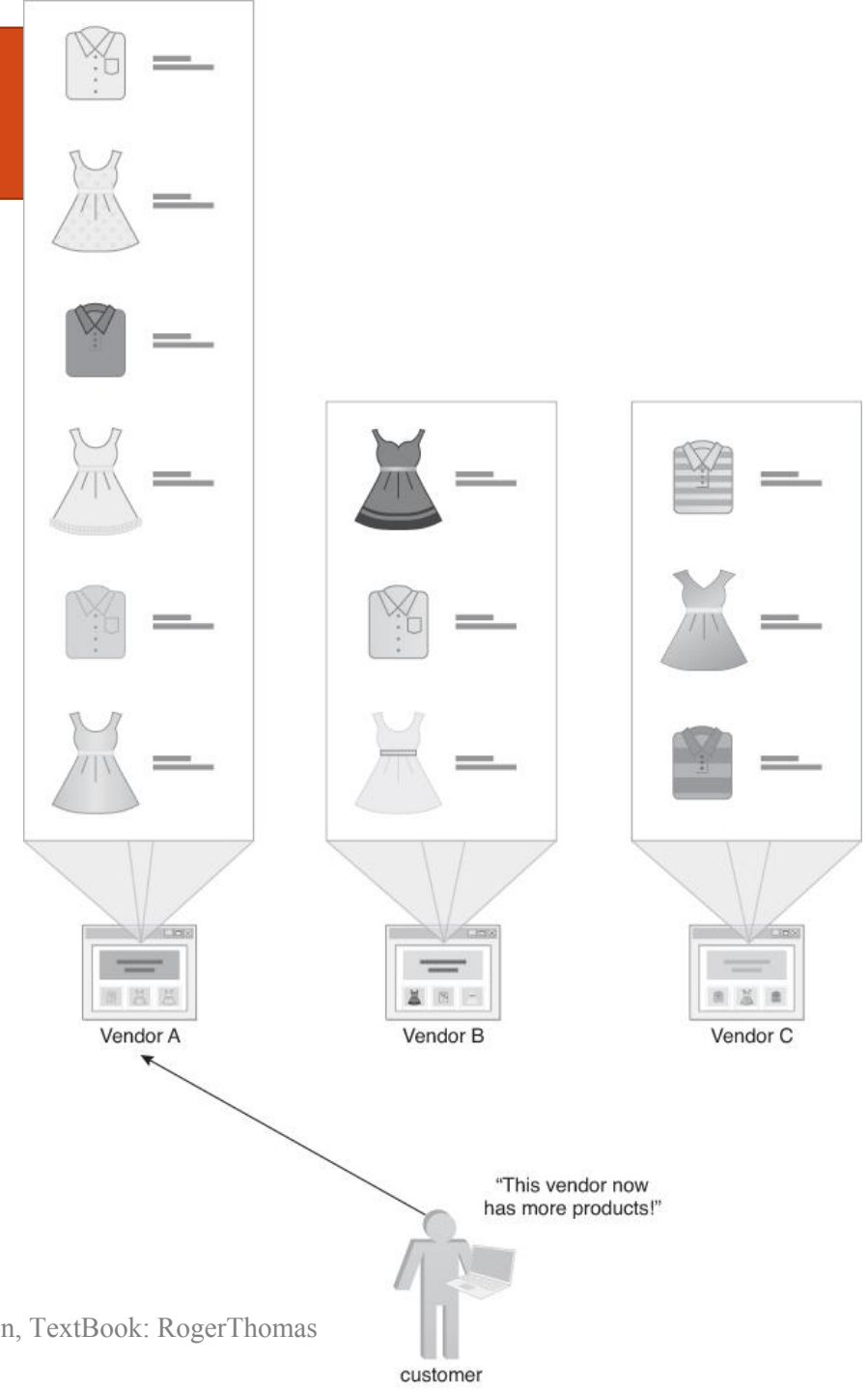
The enhanced data intelligence used in digital transformation solutions and the manner in which such solutions are designed with the primary goals of improving customer experience and introducing new innovations can help organizations break away from (or break into) a stale marketplace to noticeably distinguish themselves from competitors.

Inability to Adapt to Rapidly Changing Marketplaces

In more volatile marketplaces, change is a constant factor. Vendors may aggressively introduce new, unique products or services or impose new promotions that are difficult to compete with. Organizations in such business climates can find themselves outmatched, especially when their competitors are larger and wealthier.

Other factors that can introduce rapid change into marketplaces can be political or regulatory influences that can affect international trading or currencies. These too can lead to volatile marketplaces.

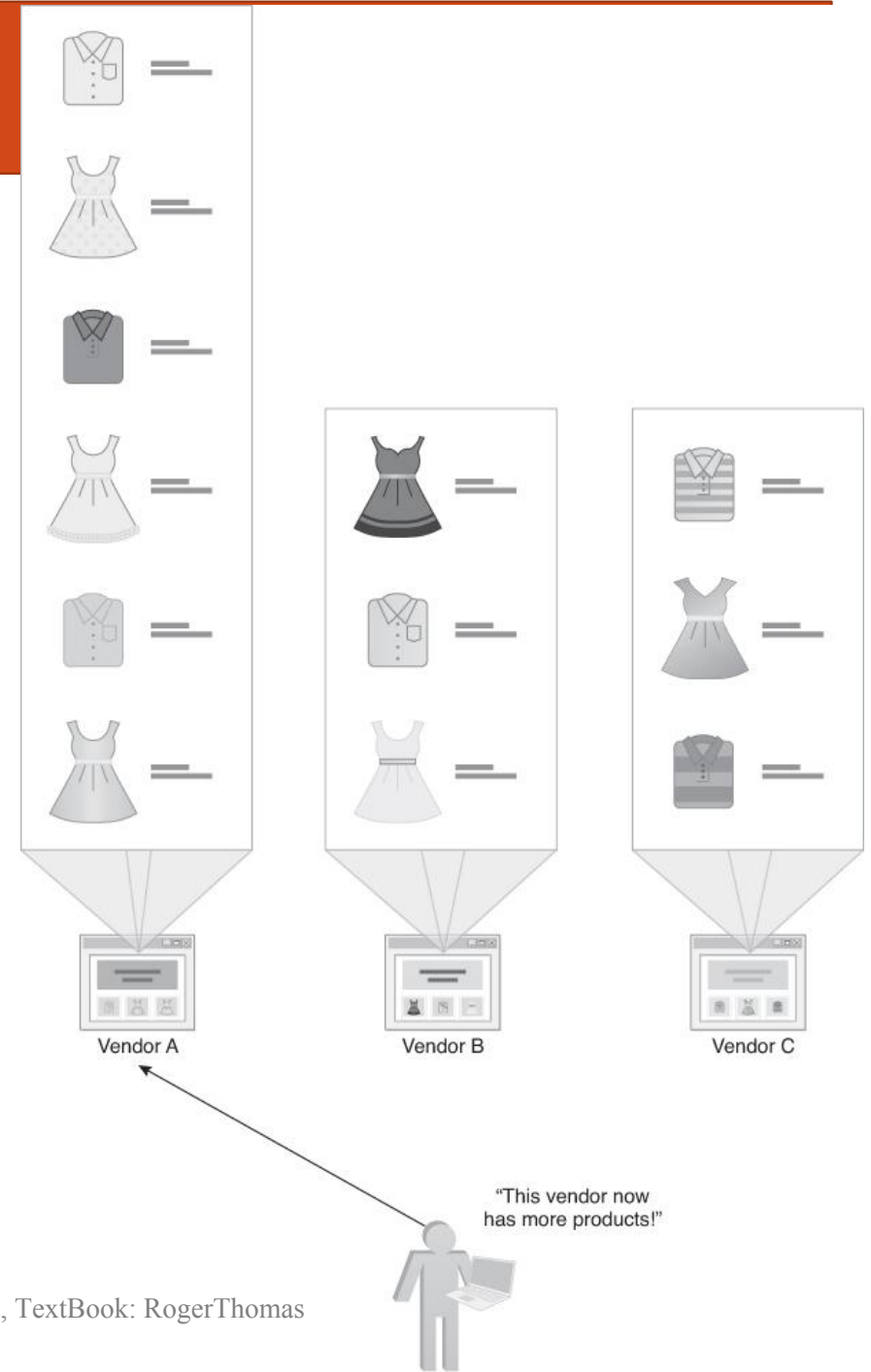
As international communities become more connected and responsive, corresponding marketplaces may be subject to greater and more rapid change. Organizations without the ability to adapt to such changes can find themselves left behind ([Figure 2.3](#)).



Digital transformation can intrinsically improve an organization's ability to act more responsively to unforeseen change, such as rapid developments in business marketplaces. Digital transformation especially supports aggressive innovation, such as to enable organizations to introduce new, disruptive products and services into a marketplace.

Figure 2.3

While Vendors B and C offer the range of products they normally do, Vendor A has begun innovating by offering a wider range of new products, making it difficult for the other vendors to compete.



Cold Customer Relationships

On a fundamental level, many organizations establish relatively impersonal relationships with their customers. Customer dissatisfaction has remained common in every industry for decades ([Figure 2.4](#)).

Many private and public organizations have recognized that there is much benefit to building “warm,” long-lasting relationships with their client communities.

However, though the desire has been there, there has traditionally been little innovation in this field that has supported the development of truly customer-centric automation solutions.

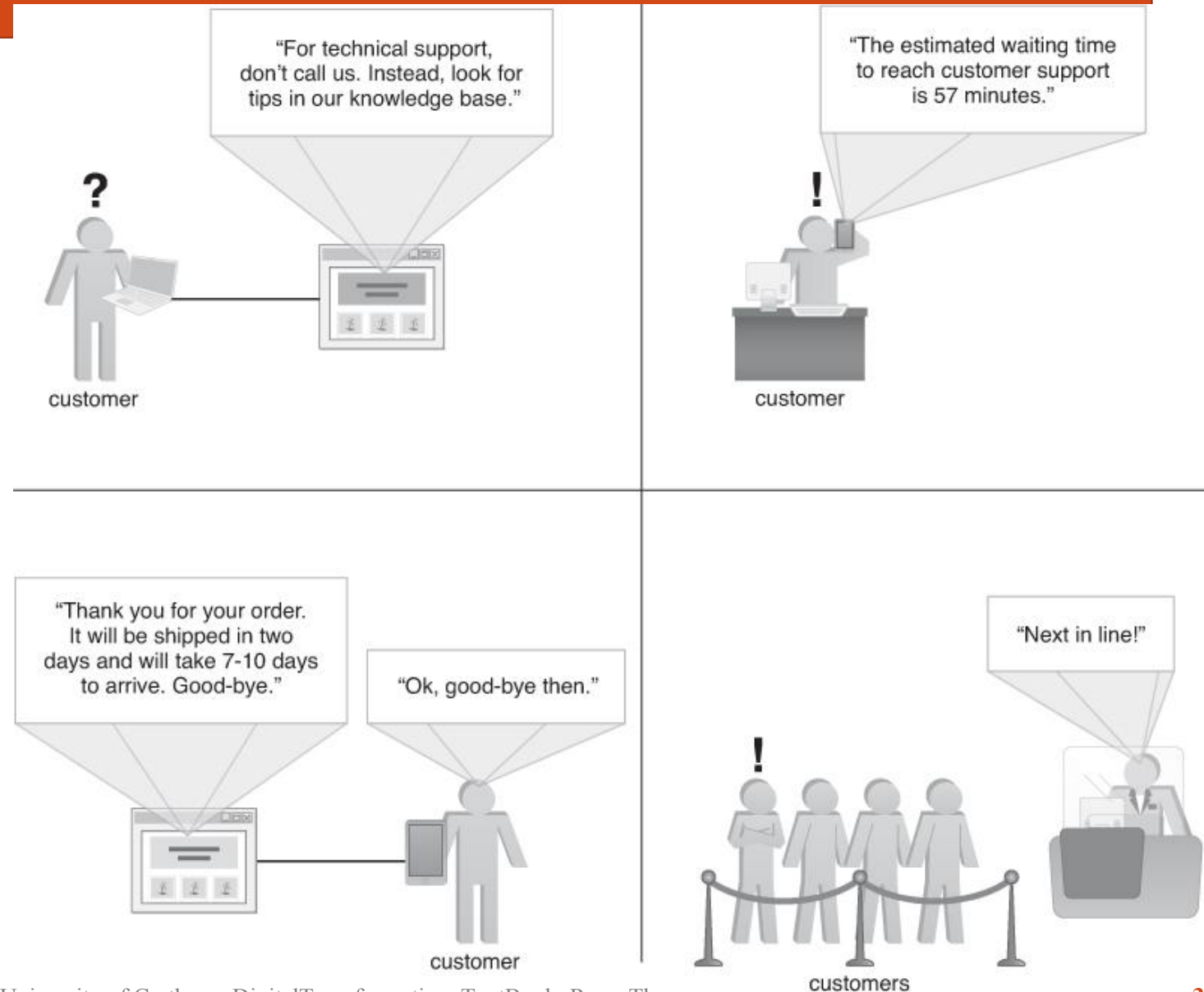
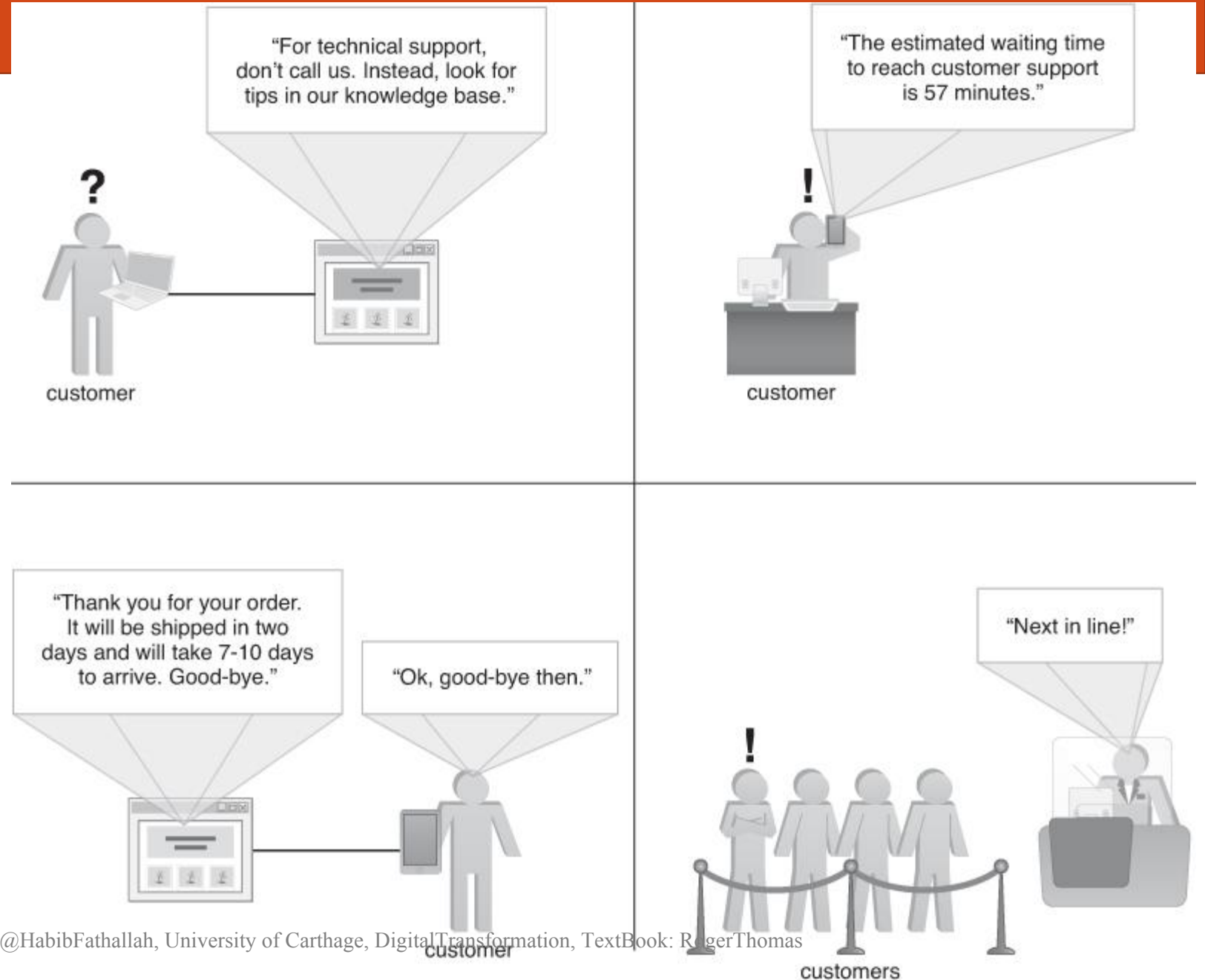


Figure 2.4:

Traditional ways in which businesses interact with customers can be relatively “cold” in that little or no effort is made to improve the customer experience.

Digital transformation provides a genuine opportunity for an organization to dramatically improve customer satisfaction for the purposes of building longer term relationships with existing customers and attracting new customers.



Inefficient Operations

While most organizations already utilize business automation technology to automate various parts of their operations, the extent of automation is often limited to those tasks that are obvious automation candidates and for which the development of automation systems is relatively straight-forward.

Many menial steps in modern operational workflows therefore remain dependent on human involvement, which can result in on-going performance limitations ([Figure 2.5](#)).

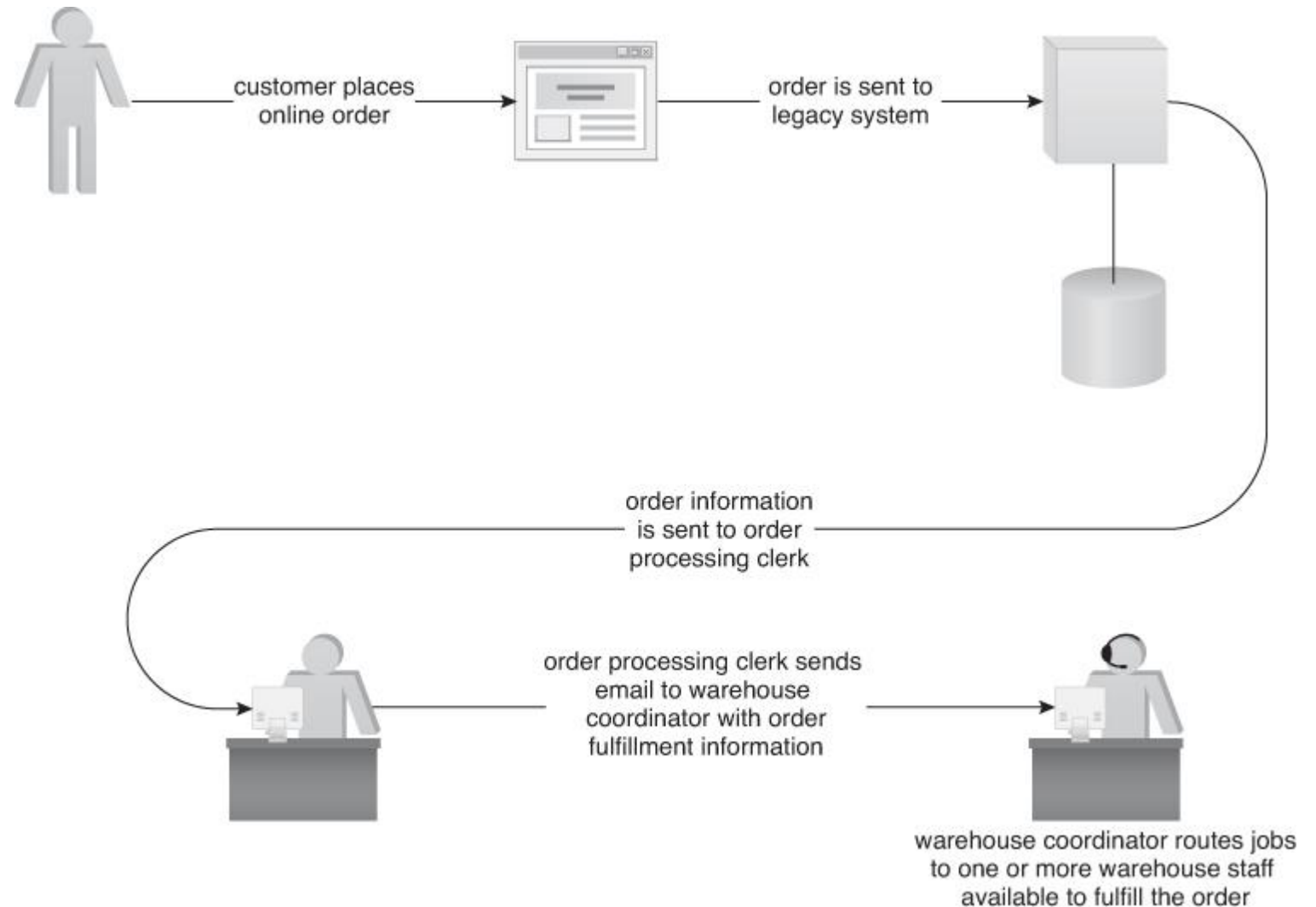
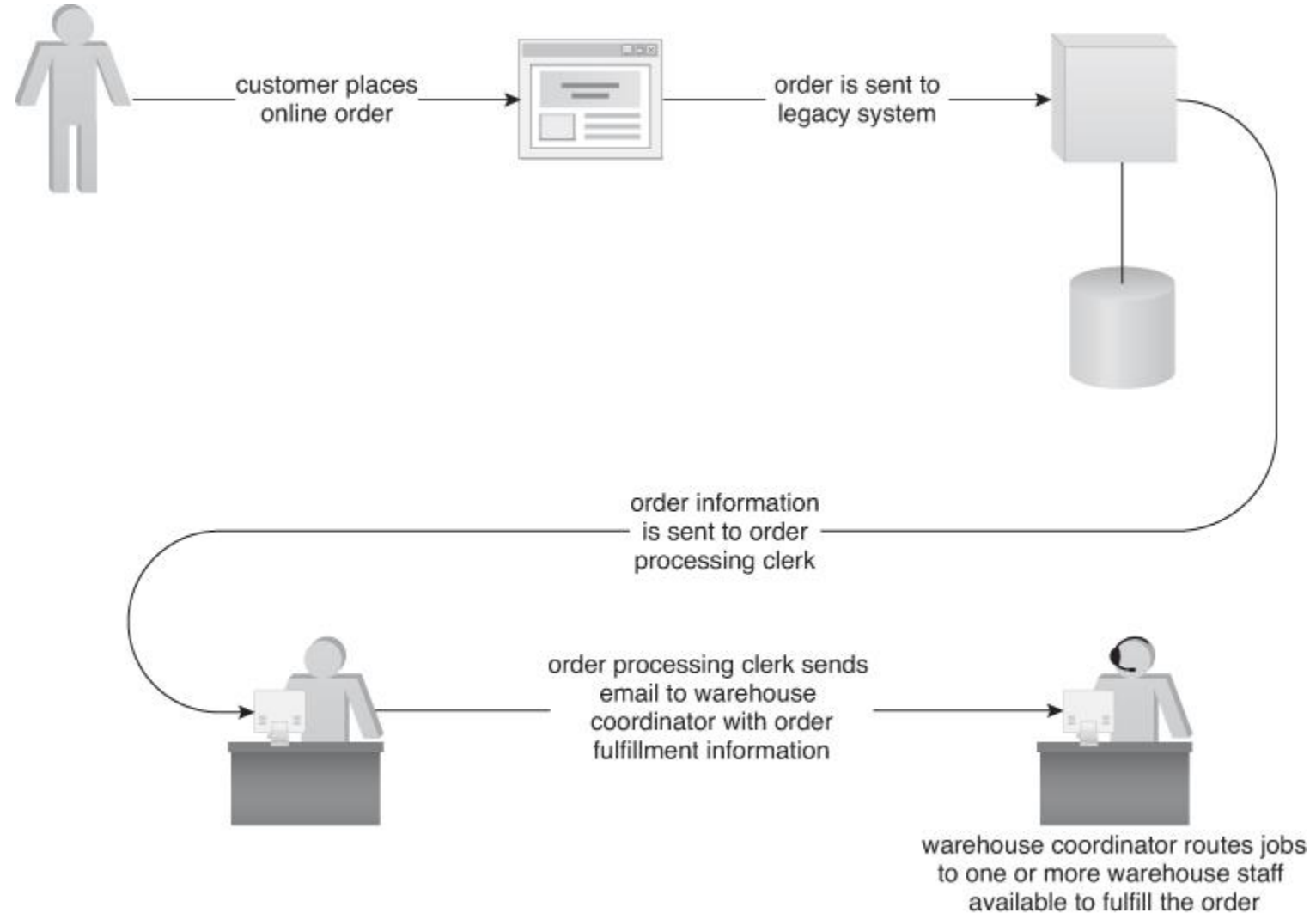


Figure 2.5

Although the collection and recording of customer orders are automated, the subsequent processing and order fulfillment assignment tasks continue to be carried out manually.

Digital transformation solutions can introduce technologies that can more easily and effectively automate tasks that have been commonly carried out manually.



Inefficient Decision-Making

Even for organizations that already have significant automation solutions powering their day-to-day operations, they traditionally continue to rely upon key personnel to make the majority of operational decisions.

Many repetitive decisions made by humans are based on simple decision-making criteria. Other decisions made by humans are based on decision-making criteria too complex for a human to process efficiently.

In either case, the involvement of a human decision maker can lead to unnecessarily inefficient operations ([Figure 2.6](#)).

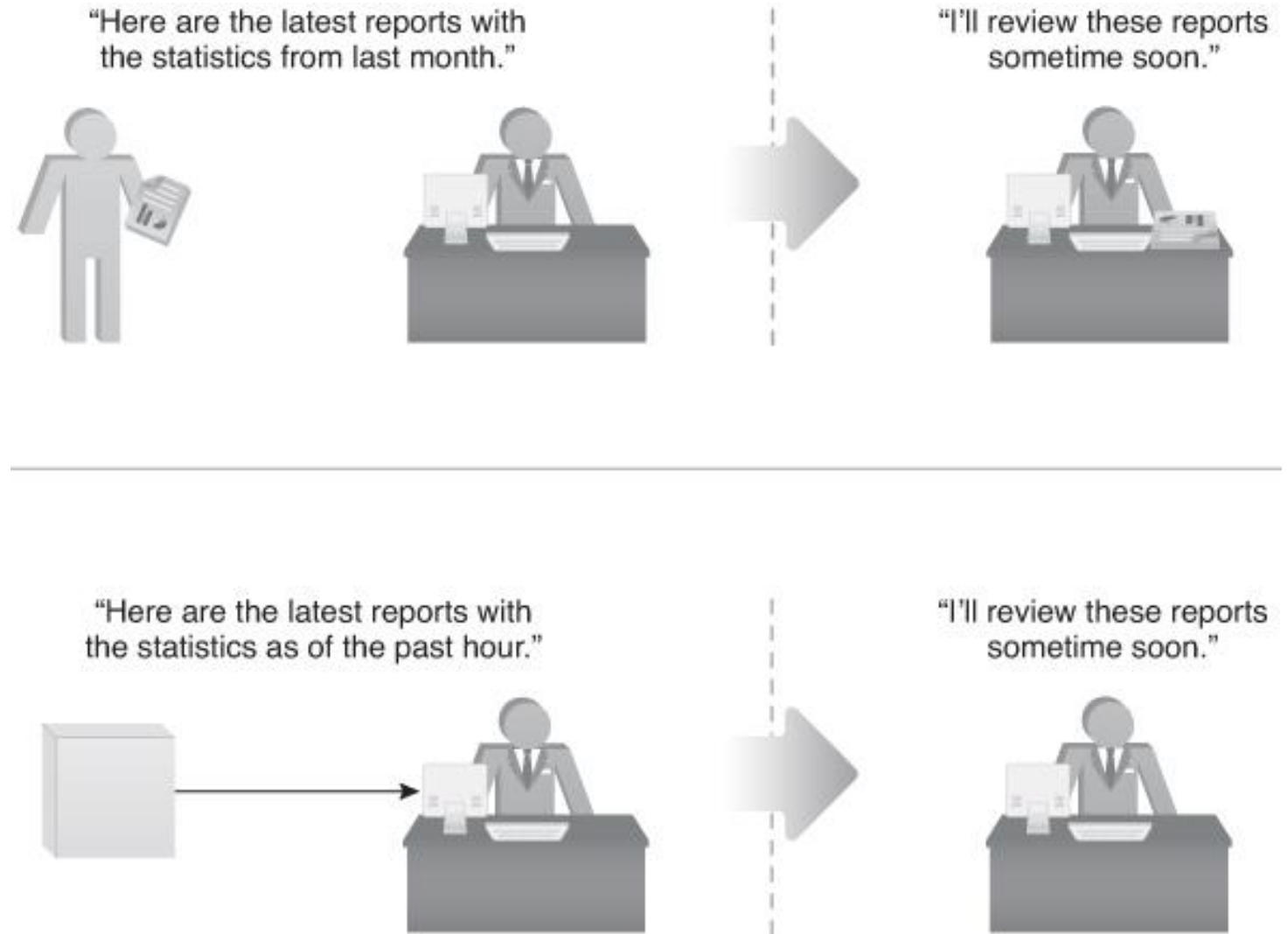
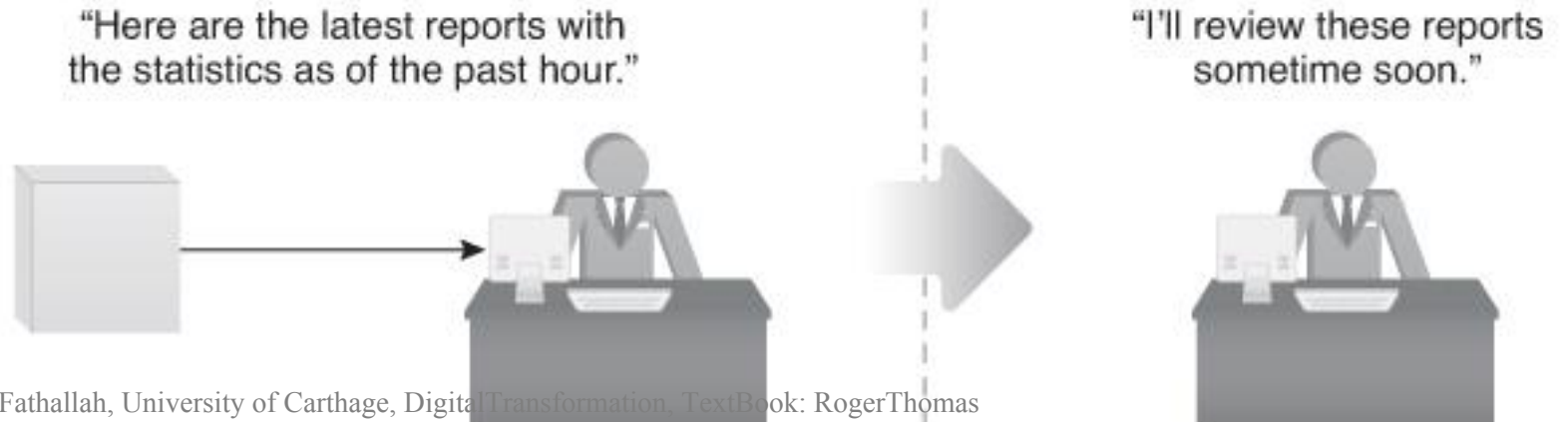
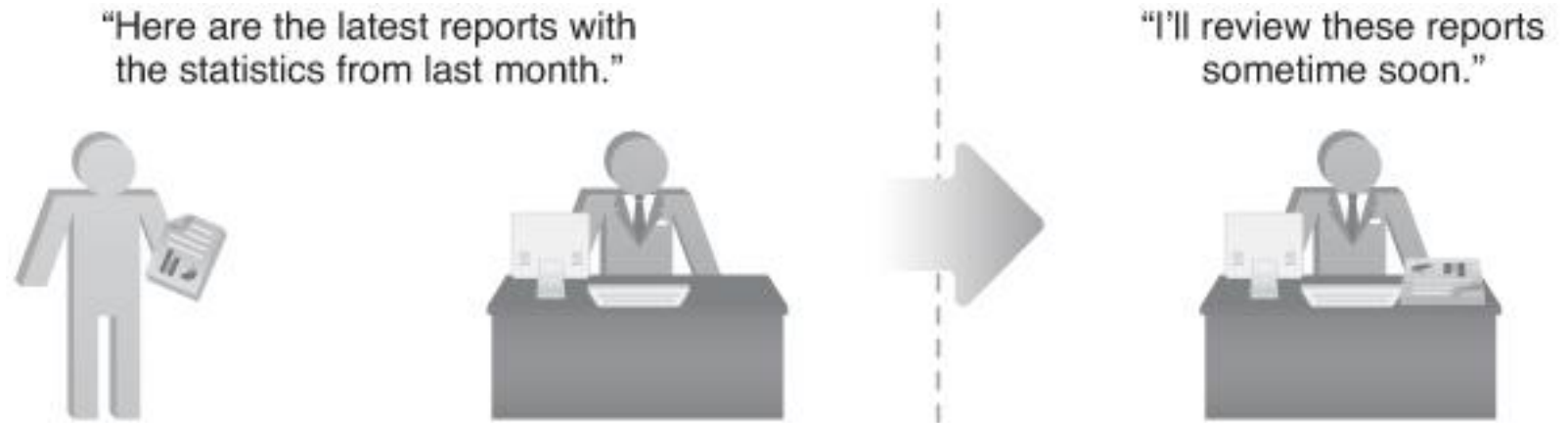


Figure 2.6

The rate at which reporting data is provided to decision makers and the time it takes humans to make decisions can be insufficiently slow to keep up with market demands.

Digital transformation platforms introduce a sophisticated level of data intelligence, as well as data-driven systems that can use this data intelligence to automate a range of decisions that have customarily been carried out by humans.



Chapter 3. Common Technology Drivers (What Enables Digital Transformation?)

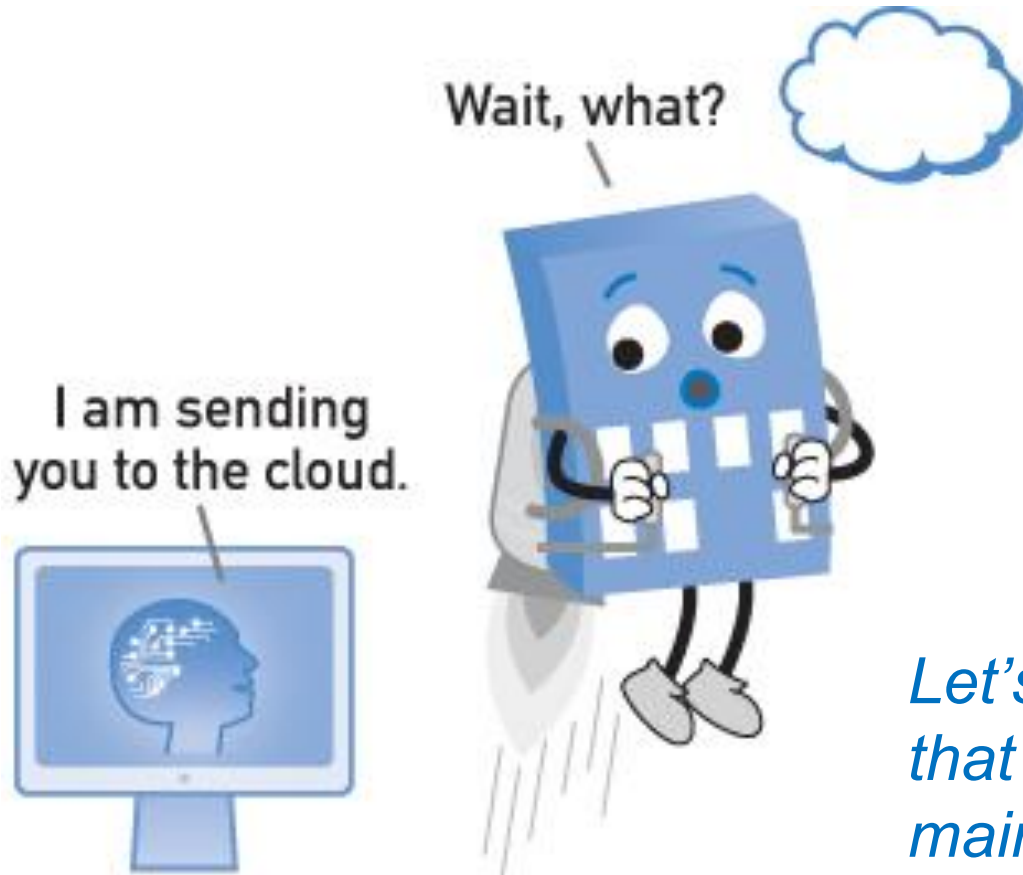
- Enhanced and Diverse Data Collection
- Contemporary Data Science
- Sophisticated Automation Technology
- Autonomous Decision-Making
- Centralized, Scalable, Resilient IT Resources
- Immutable Data Storage
- Ubiquitous Multiexperience Access



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Table of Content:

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Let's now highlight some of the useful technologies that drove digital transformation into the IT mainstream. Organizations often need an open mind to fully comprehend how and to what extent the capabilities provided by these technologies can support their business goals.

As established in [Chapter 1](#), the availability of a specific set of technology innovations has made digital transformation possible. This chapter explains how these technologies have acted as the primary enablers of digital transformation.

The following key technology drivers are covered:

- ❑ Enhanced and Diverse Data Collection
- ❑ Contemporary Data Science
- ❑ Sophisticated Automation Technology
- ❑ Autonomous Decision-Making
- ❑ Centralized, Scalable, Resilient IT Resources
- ❑ Immutable Data Storage
- ❑ Ubiquitous Multiexperience Access

Enhanced and Diverse Data Collection

Digital transformation platforms encompass technology innovations capable of collecting an extremely broad range of diverse data.

For example:

- ❑ Customer-centric solution design and associated monitoring technology can capture a great deal of customer profile and behavioral data from how and when customers interact with an organization's existing applications.
- ❑ Solutions can encompass a network of remote sensors capable of collecting and streaming large quantities of telemetry data from the outside world, including human activity, vehicle activity, transport logistics, weather, vibration and many activities associated with electronic devices.
- ❑ Third-party market data can be acquired and processed through data analytics solutions that can extract meaningful insights into individuals and entire communities.
- ❑ Paper-based documents can be digitized to provide a greater archive of input data which the organization can further subject to analysis and processing in conjunction with other input data.

By having such a wide range of data as input, digital transformation platforms can:

- ❑ offer human decision makers deeply insightful data intelligence to help improve their decision-making capabilities and to further help them discover how new product and service innovations could be introduced
- ❑ perform highly sophisticated and intelligent forms of automation that can improve both the quality and efficiency of how business is automated ([Figure 3.1](#))
- ❑ enable the opportunity to automate decision-making responsibilities as part of digital transformation automation solutions (as discussed separately in [Chapter 8: Intelligent Decision-Making](#))

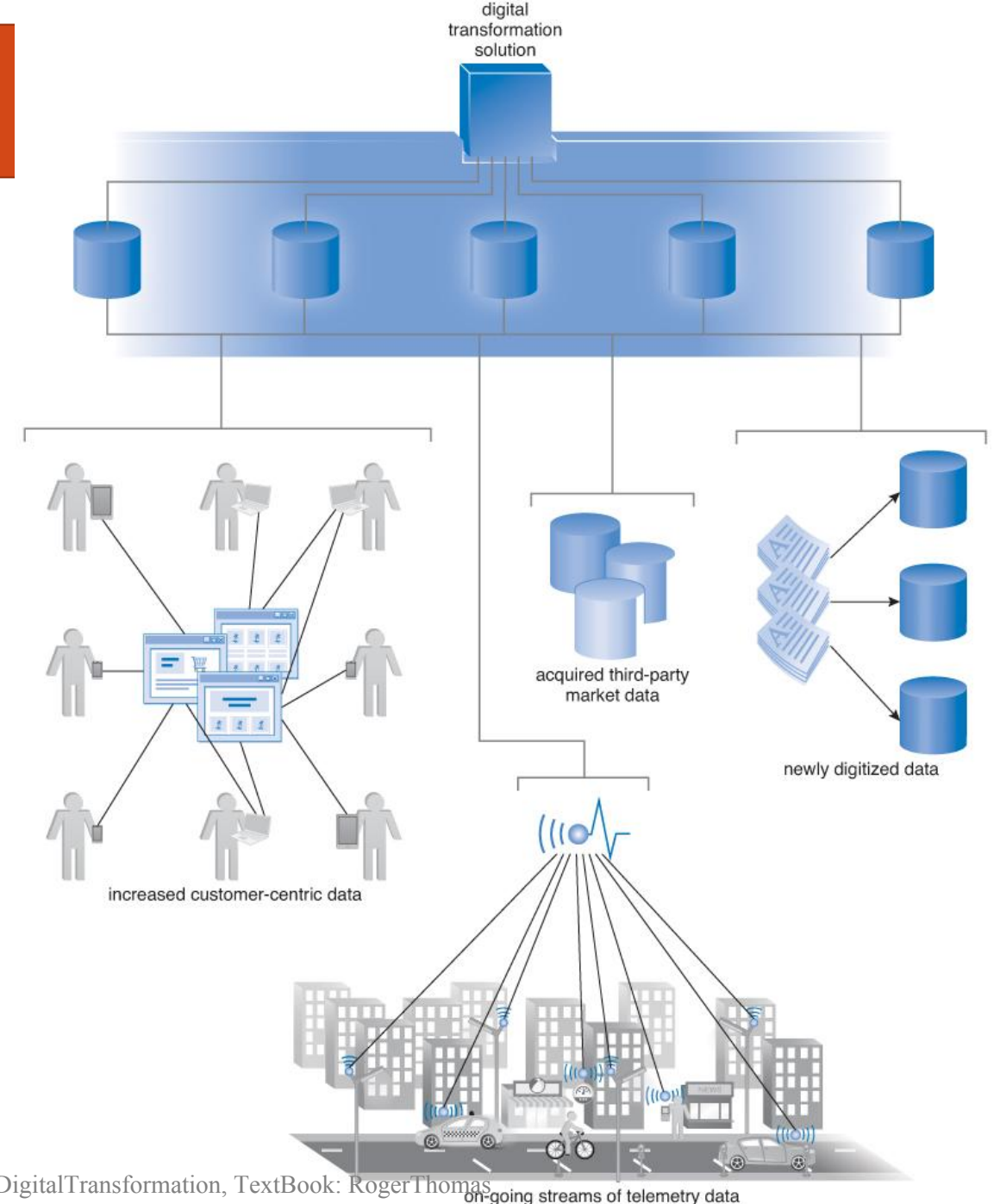
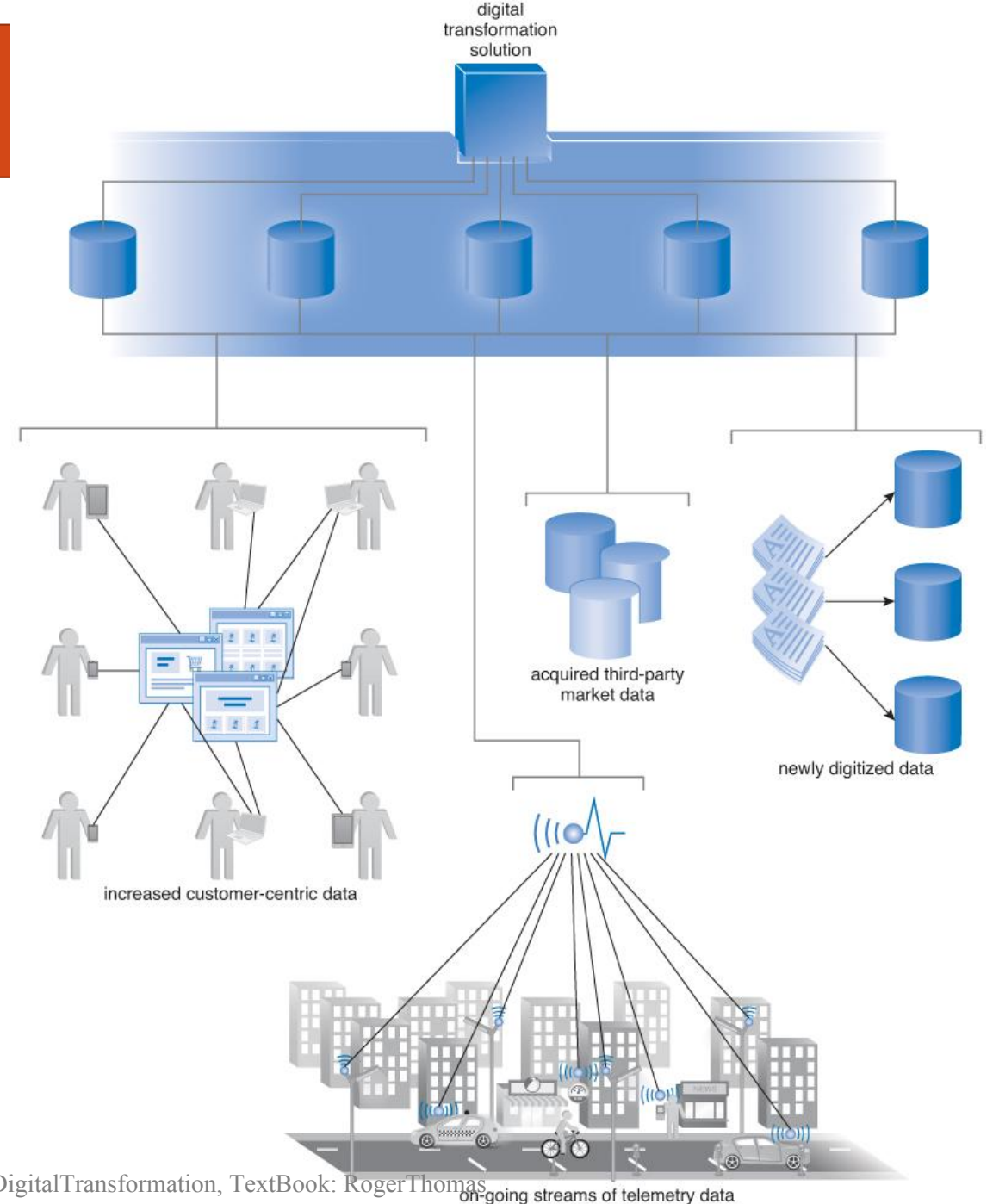


Figure 3.1

A digital transformation solution may access and utilize data that was collected from a wide variety of sources. This enables the solution to perform its functions more effectively and to further improve how it performs its functions over time.

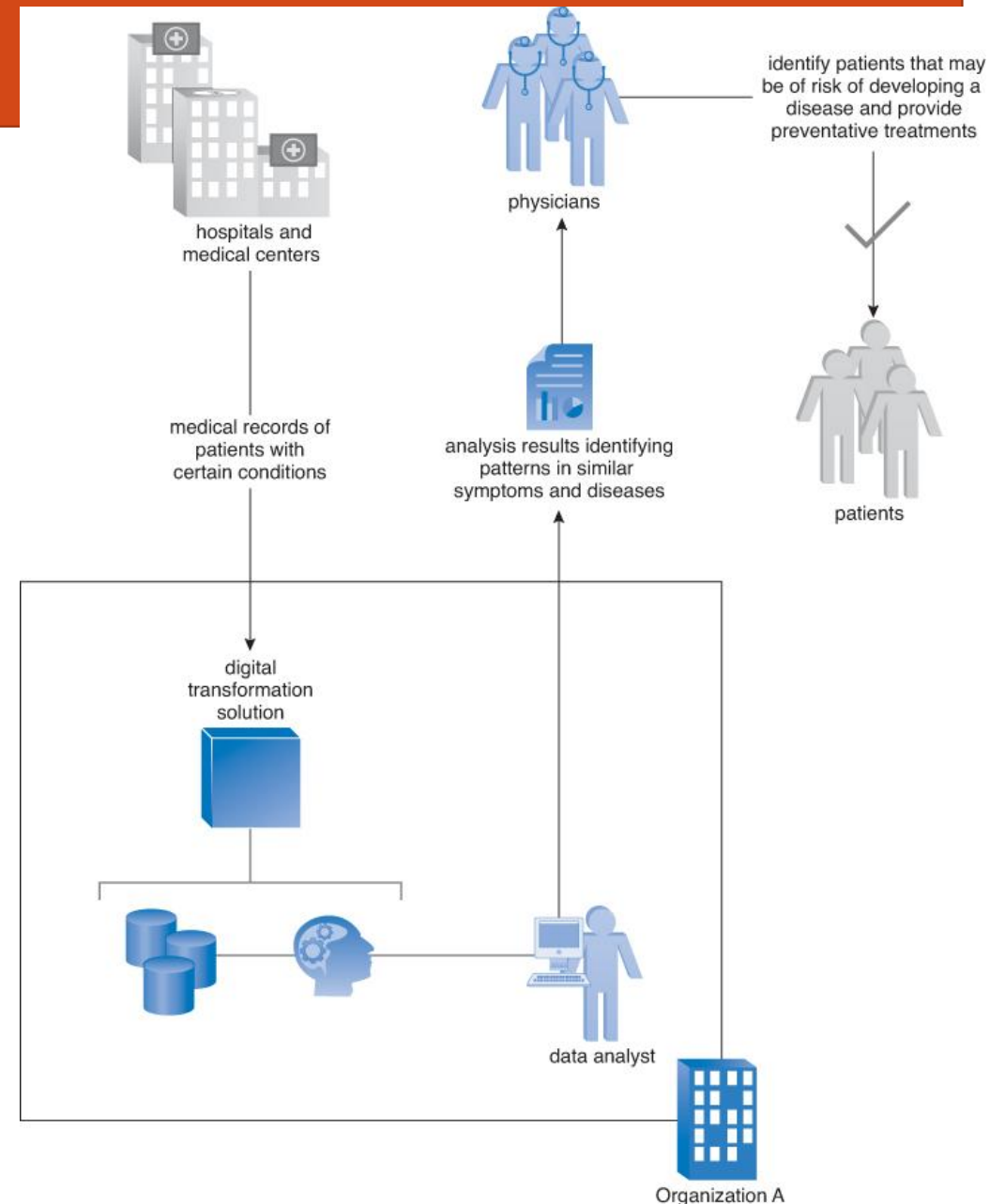


Technologies relevant to this driver include:

- Big Data Analysis and Analytics
- Internet of Things (IoT)
- Robotic Process Automation (RPA)

Figure 3.2

Medical records provided by hospitals and medical centers are used by physicians to help identify patterns that assist in diagnosing patients with similar conditions or predicting the likelihood of such conditions developing



Sophisticated Automation Technology

Manual tasks that have traditionally not been easily automated can now be fully automated. Special software programs called “bots” can be configured to carry out a range of hands-on processing tasks, such as data entry and information routing ([Figure 3.3](#)). The behavior and functionality of bots can be further enhanced through the use of data science technologies.

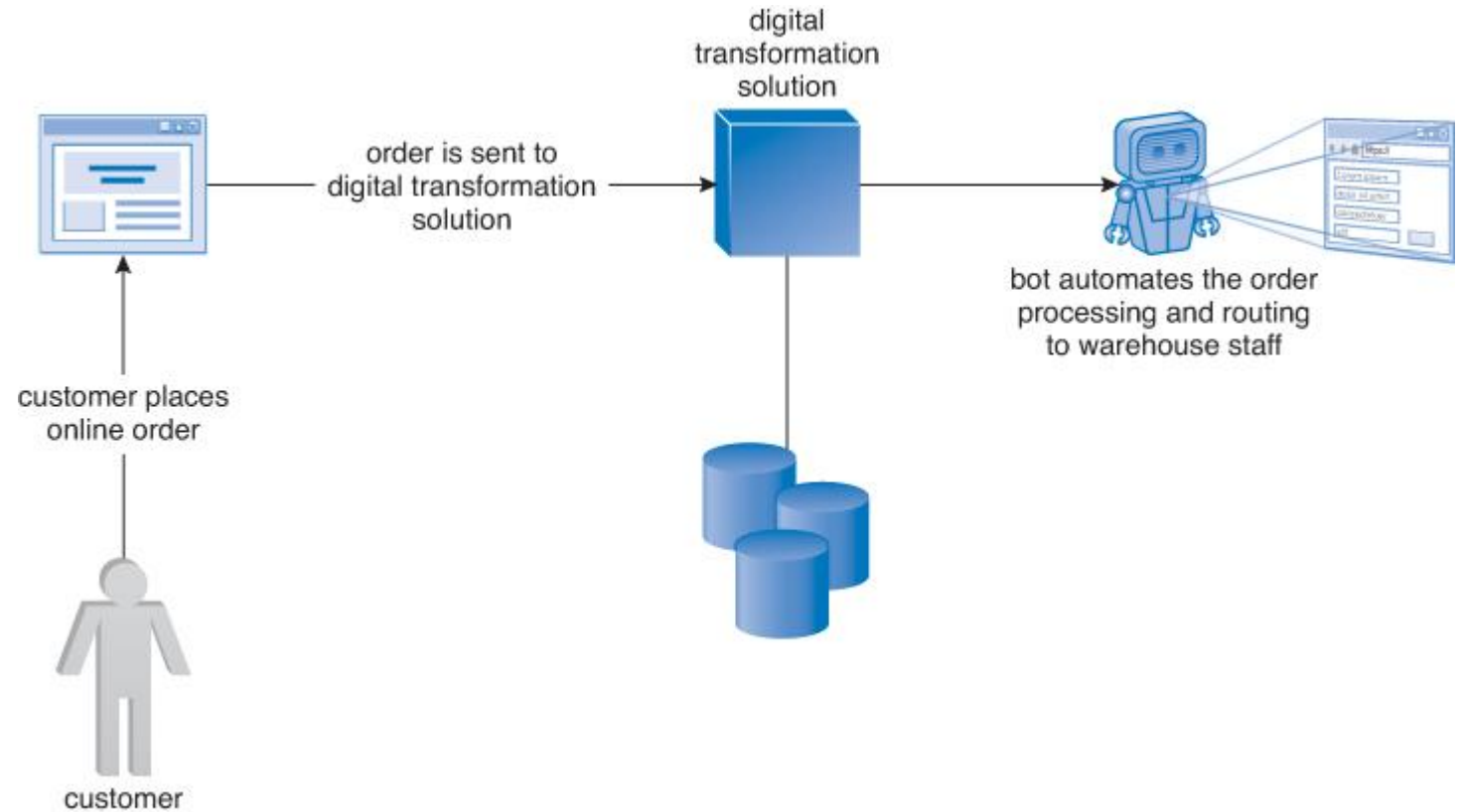


Figure 3.3

A software program known as a “bot” is given the responsibility to complete order processing-related tasks that were previously carried out by humans.

Autonomous Decision-Making

Modern digital transformation solutions can incorporate data science technologies that can assume decision-making responsibilities. This enables the solution to rapidly respond to a range of situations that may have previously required more time for correct human decisions to be carried out ([Figure 3.4](#)).

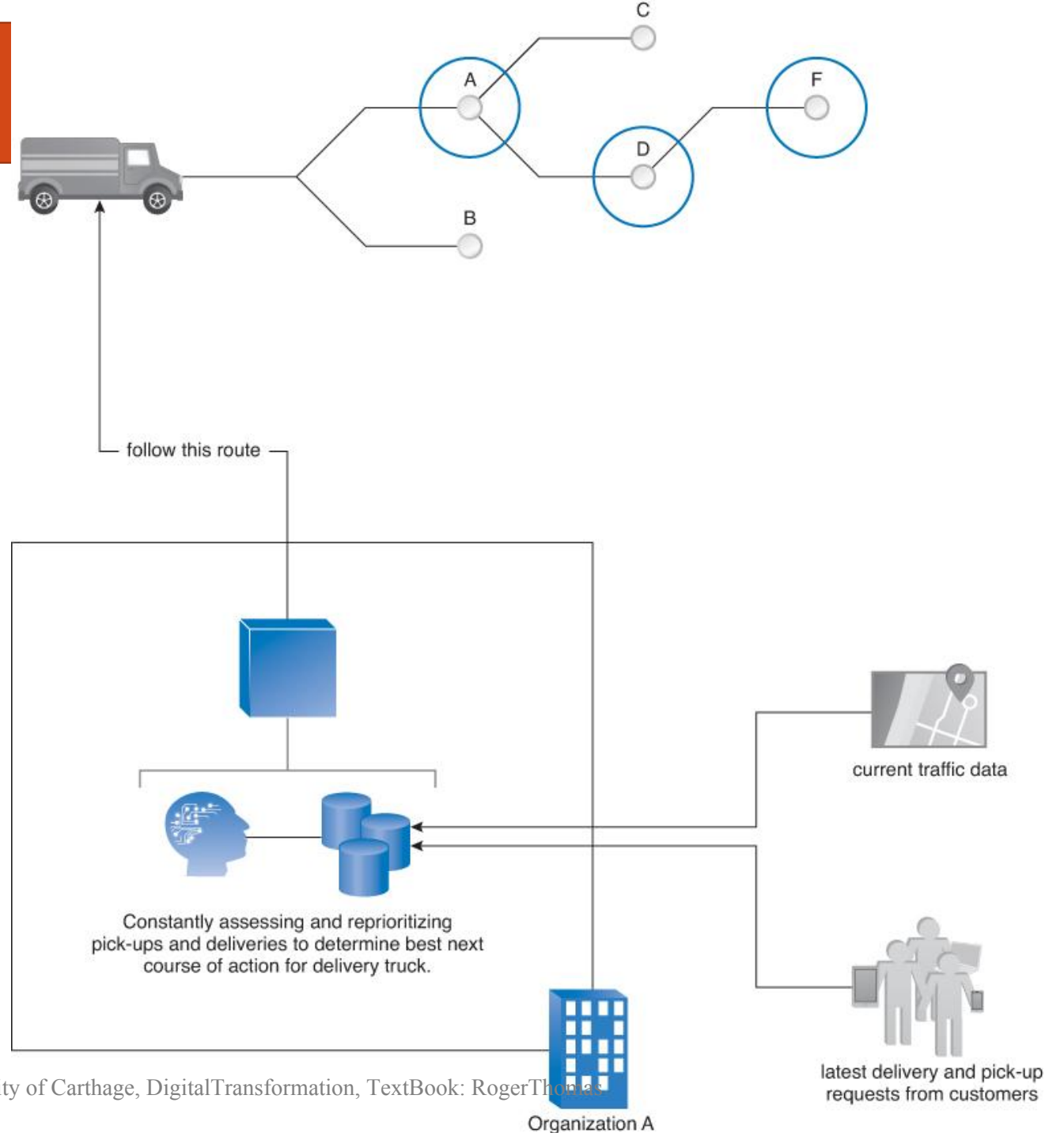
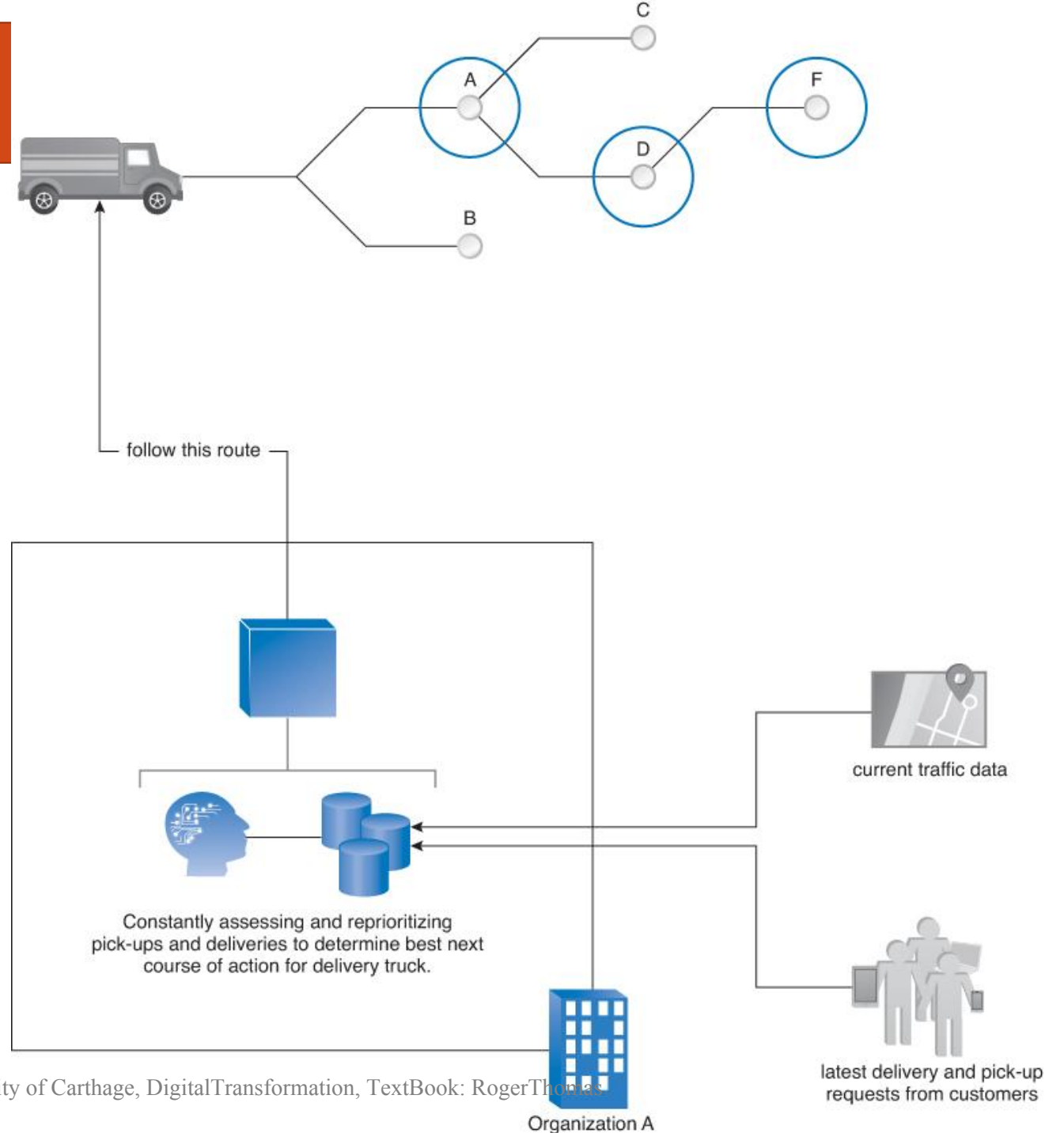


Figure 3.4

A digital transformation solution dynamically determines the route a delivery truck should take to visit the warehouses where deliveries are to be transported. The system takes into account current traffic data and current customer request data.



Centralized, Scalable, Resilient IT Resources

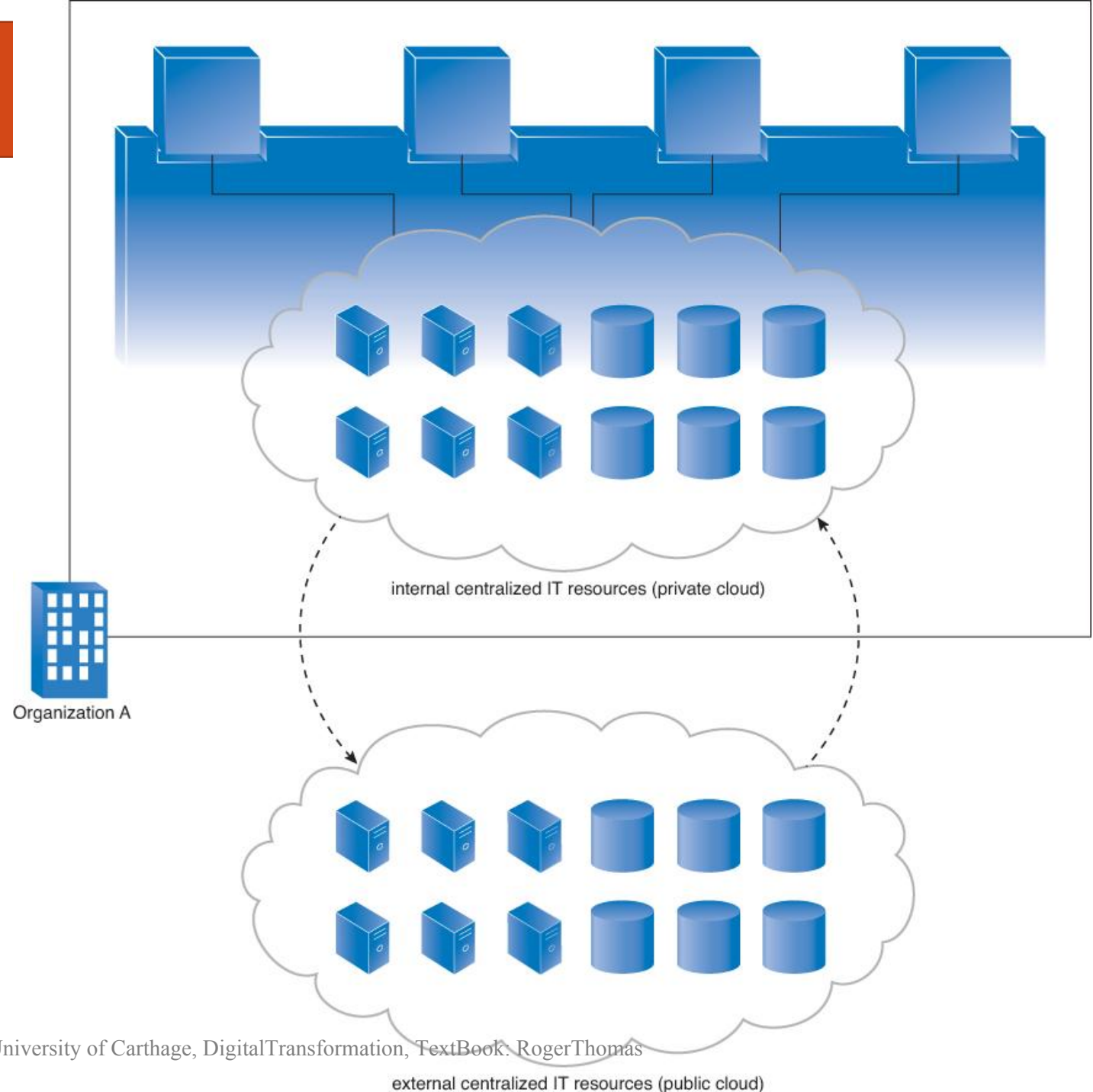
Digital transformation solutions commonly have distinct performance requirements, such as:

- ❑ high scalability to accommodate large quantities of concurrent users
- ❑ dynamic scalability to accommodate unpredictable usage fluctuations
- ❑ high resiliency to recover from failure with minimal disruptions in service
- ❑ high compute power to process large quantities of data
- ❑ rapid provisioning of new solutions and solution updates
- ❑ centralized IT resources that can be shared by multiple solutions

Cloud-based environments can provide the necessary infrastructure to support these and other digital transformation solution performance requirements.

Such environments can be established within the organization or they can be leased from third-party providers.

Organizations can also have their own private cloud that can automatically scale out to an external third-party cloud, when the processing limitations of the private cloud are exceeded ([Figure 3.5](#)).



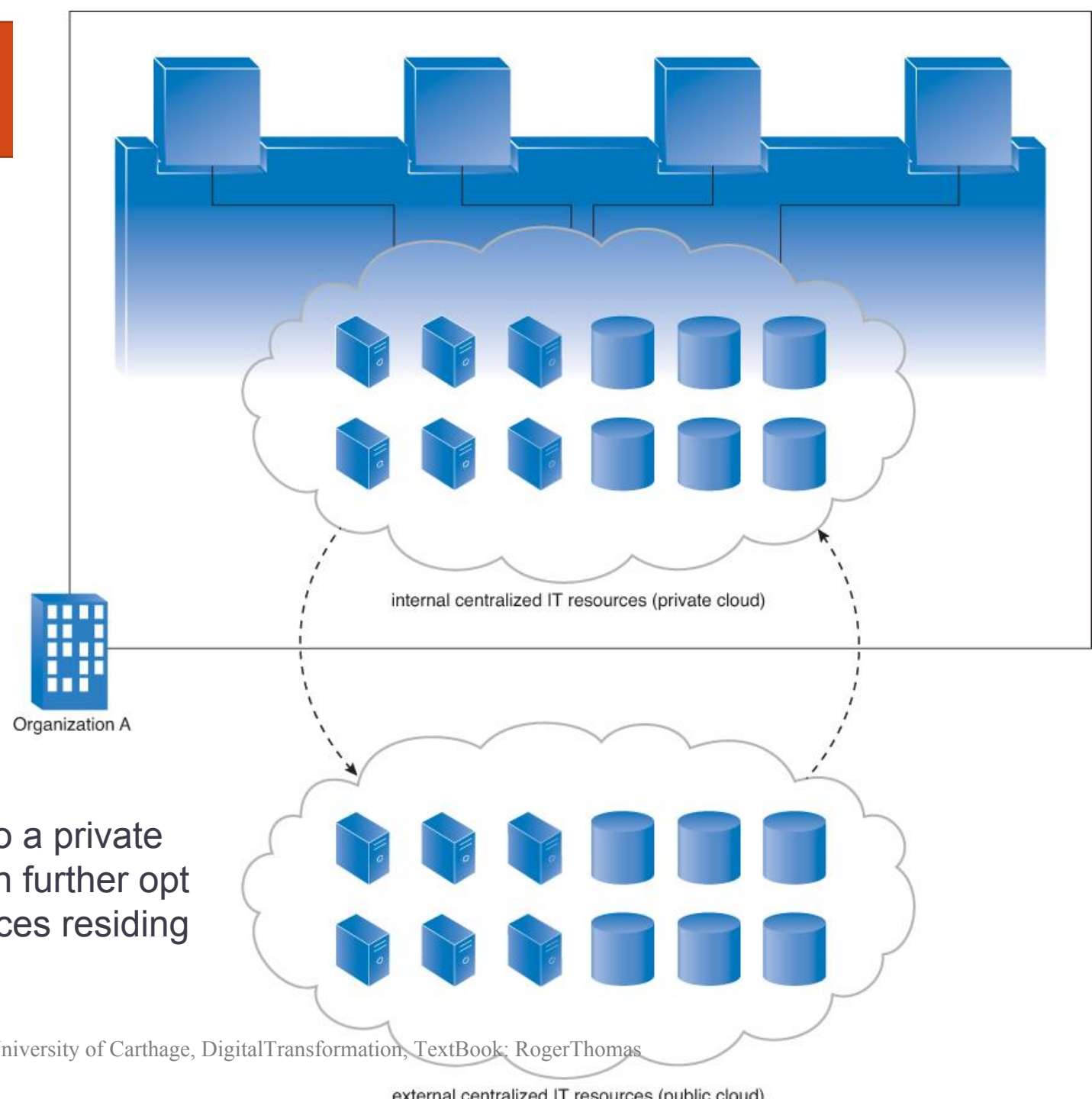


Figure 3.5

An organization can centralize IT resources into a private cloud that it owns and operates internally. It can further opt to balance or “burst-out” to additional IT resources residing in a public cloud, when necessary.

Immutable Data Storage

With the increase in the quantity of digital data being collected, there is a greater need for sensitive and important data to be permanently stored in such a manner that it can never be altered or manipulated.

Highly secure storage technology that emerged from the use of cryptocurrencies has made it possible to position immutable repositories alongside corporate databases, as part of digital transformation solutions ([Figure 3.6](#)).

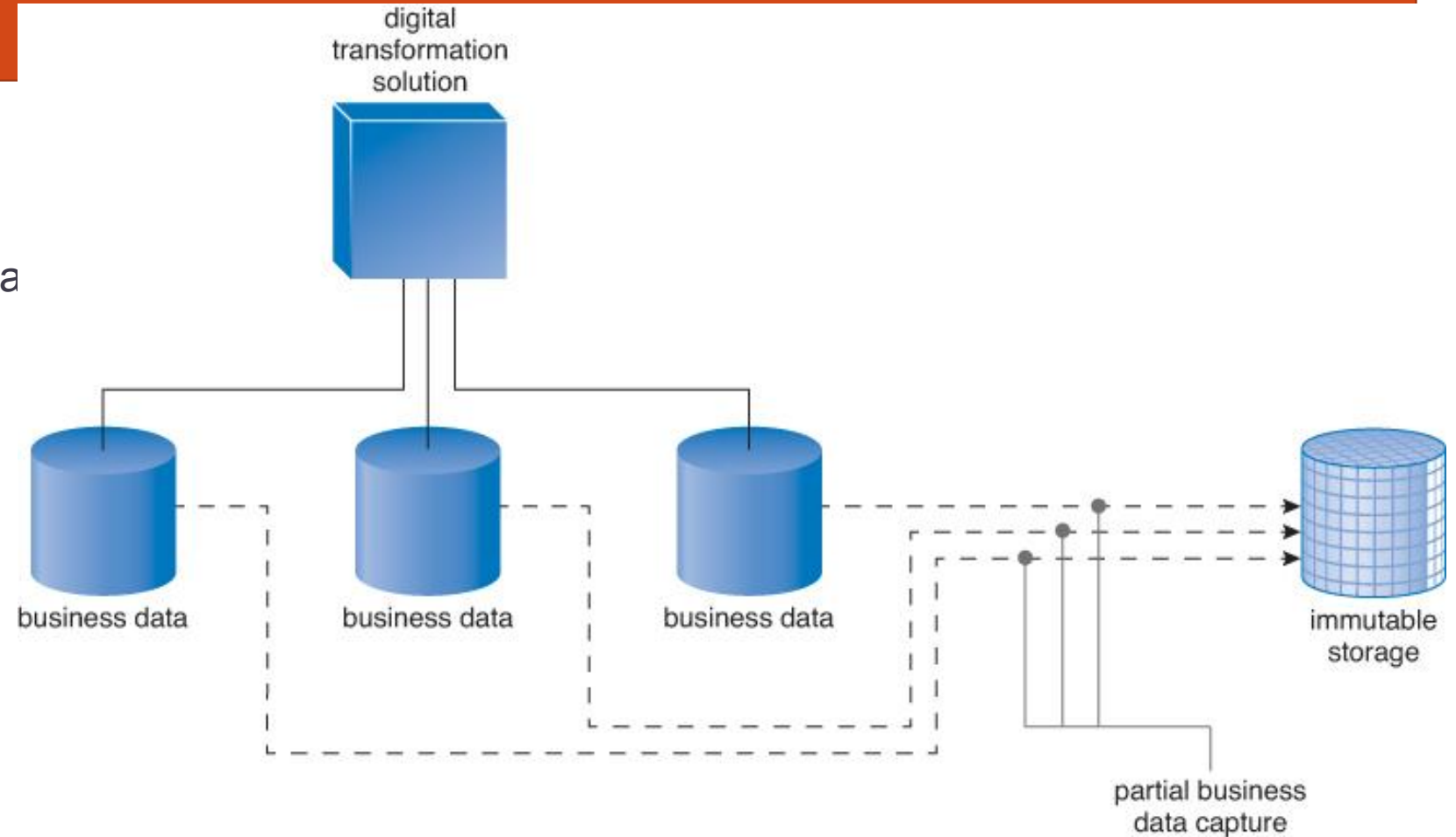


Figure 3.6

A digital transformation solution submits important business records to a repository capable of immutable storage.

Ubiquitous Multiexperience Access

A key objective when designing customer-centric digital transformation solutions is to provide “multiexperience” support by enabling ubiquitous access to a range of business interaction options (or *channels*) that that the customer can choose from ([Figure 3.7](#)). The *Single vs. Multi vs. Omni-Channel Customer Interactions* section in [Chapter 6](#) provides examples of multi experience access methods.

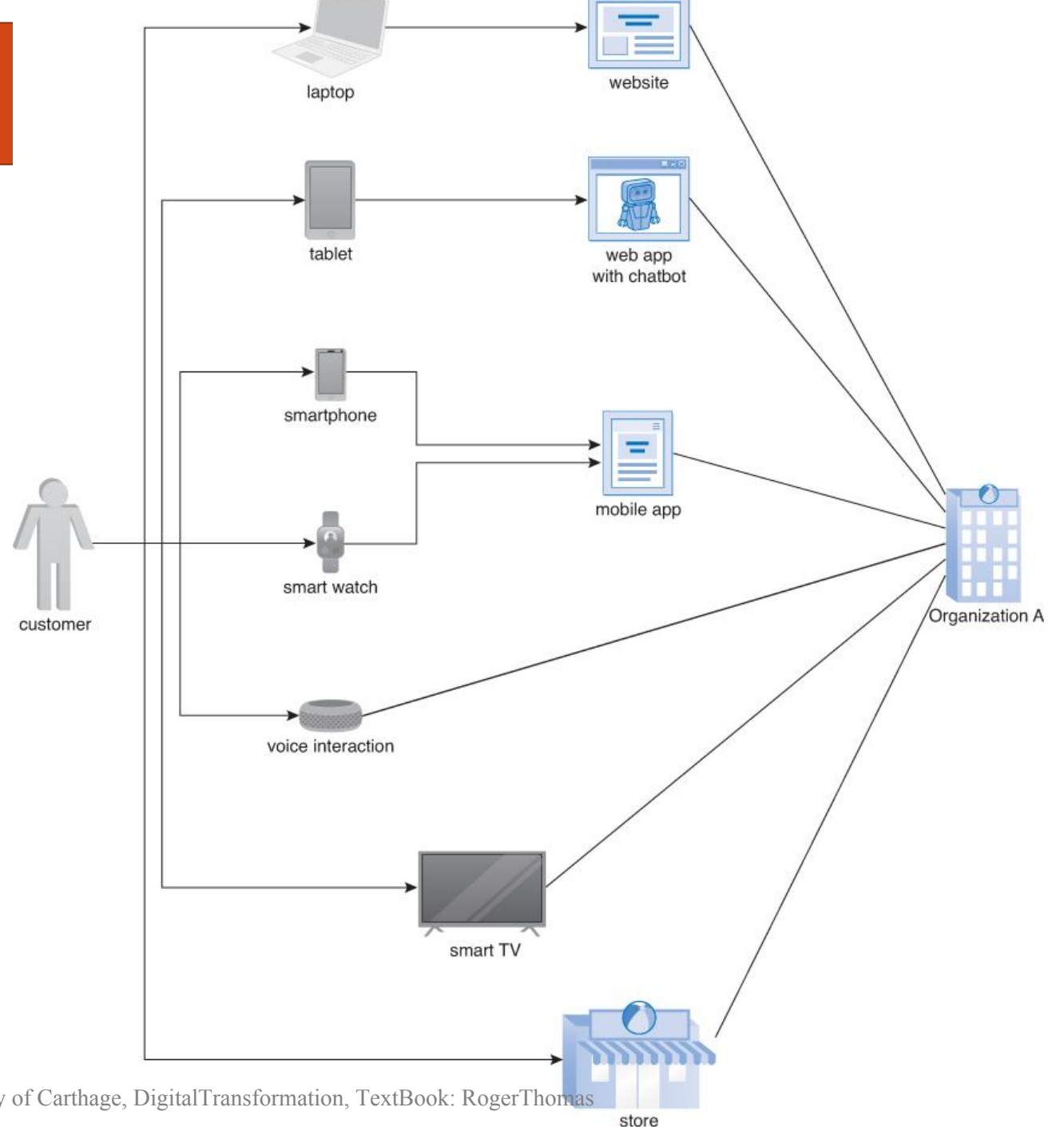
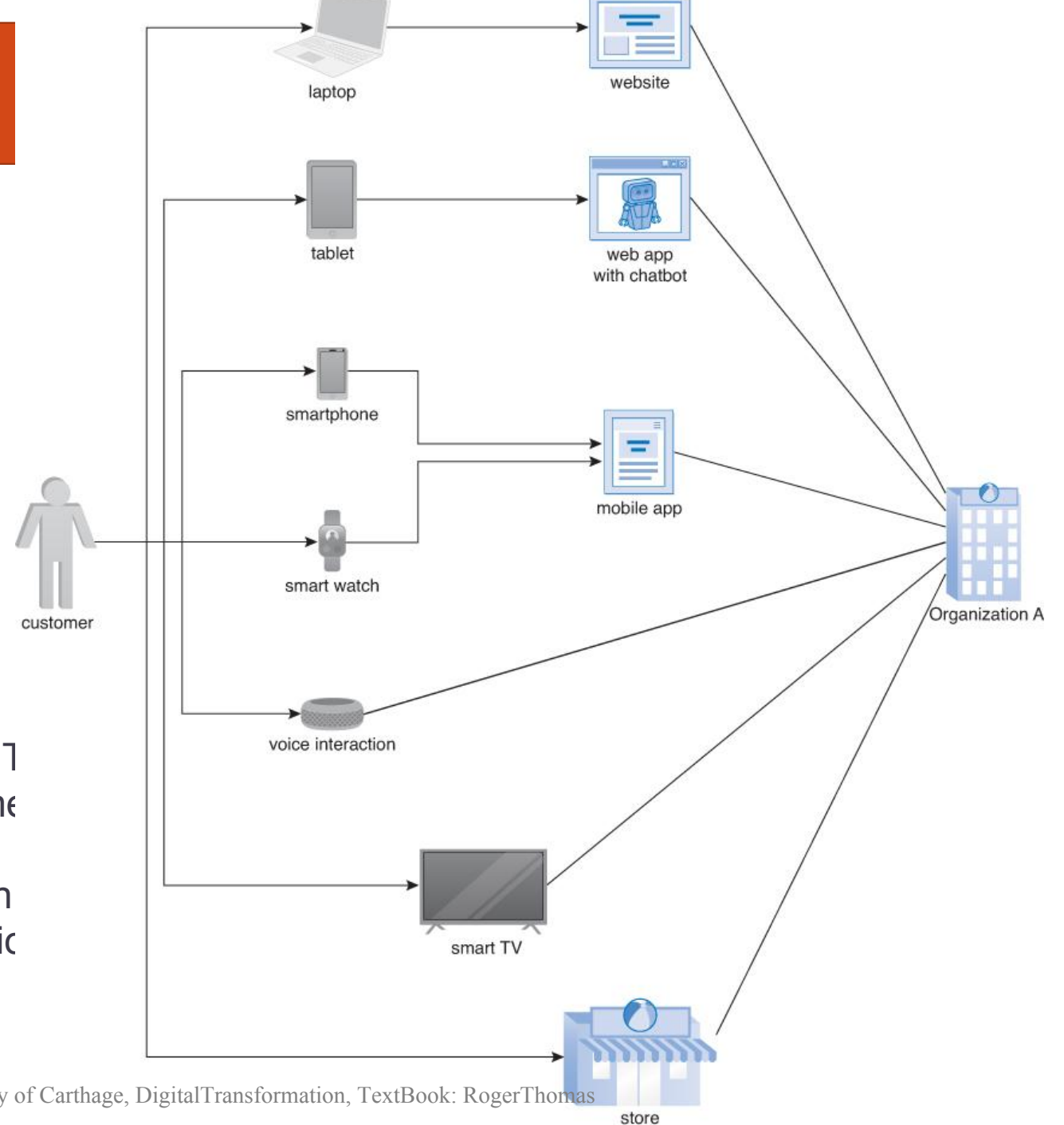




Figure 3.7

A customer may choose to interact with a business using any number of access methods or channels. The business is able to maintain the state of the customer interaction from one channel to the other.

For example, the customer might start a transaction using a mobile app and then complete the transaction when visiting the store in person.





The utilization of cloud-based environments, along with other technologies, makes it possible to support multiple access channels whereby the “state” of the customer’s activity is always preserved across the different access method types.

For example, a customer may start a transaction online using a tablet, and can then (at a later point) switch to a voice-activated device to complete the transaction.